EL3004- FEEDBACK CONTROL SYSTEMS

LAB MANUAL

DEPARTMENT OF ELECTRICAL ENGINEERING,

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List of Equipment

Sr. No.	Description
1	Computer
2	NI ELVIS II Trainer
3	QNET DC Motor Control Trainer
4	QNET Rotary Pendulum Trainer

EXPERIMENT 1
Study of first order systems

Objective:

- To design first order RL and RC circuits and Analyze their Frequency responses.

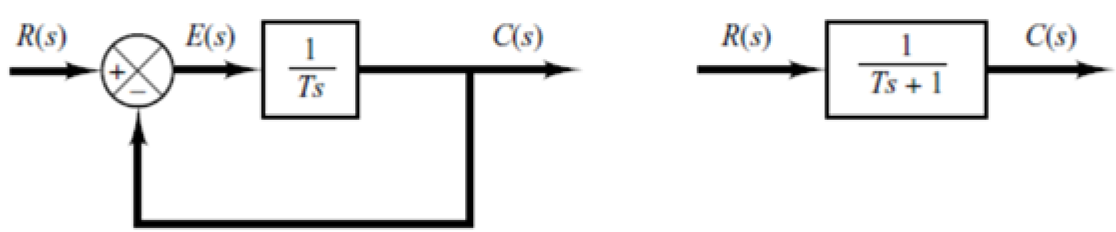
Equipment Required:

- Oscilloscope with probes
- Function generator
- Digital multi-meter
- Power supply
- Bread Board
- Capacitor
- Inductor
- Resistor

Theory:

First order systems are, by definition, systems whose input-output relationship is a first order differential equation. A first order differential equation contains a first order derivative but no derivative higher than first order – the order of a differential equation is the order of the highest order derivative present in the equation.

The first order system has only one pole as shown



(a) (b)

Figure1:(a) Block Diagram of a first order system (b) Simplified block Diagram

$$\frac{C(S)}{R(S)} = K\frac{1}{TS+1}$$

Where K is the DC gain and τ is the time constant of the system.

- Time constant is a measure of how quickly a 1st order system response to a unit step input.
- With step input, we can measure the time constant and steady state value, from which transfer function can be calculated.
- If we can identify τ and K from laboratory testing, we can obtain the transfer function of the system.

- Frequency response is characterized by the magnitude of the system's response for different frequencies typically measured in decibels(dB). A Bode diagram consist of two graphs: One is a plot of the logarithm of magnitude of a transfer function; the other is a plot of the phase angle; both are plotted against the frequency on a logarithmic scale. In the logarithmic representation, the curves are drawn on semilog-paper, using the log scale for frequency and the linear scale for either magnitude(but in decibels) or phase angle(in degrees)
- Asymptotic bode plot is a simple method for sketching log-magnitude on an approximate curve.The graph represents straight-line asymptotes which are sufficient if only rough information on the frequency-response is needed.

Part(a):Design of first order circuit

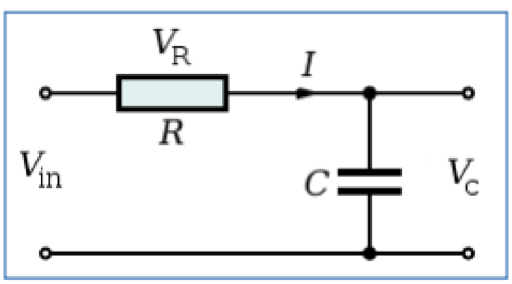


Figure2(System1)

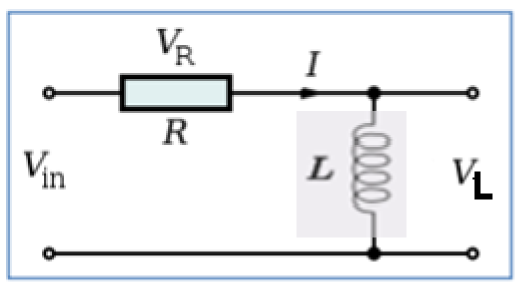
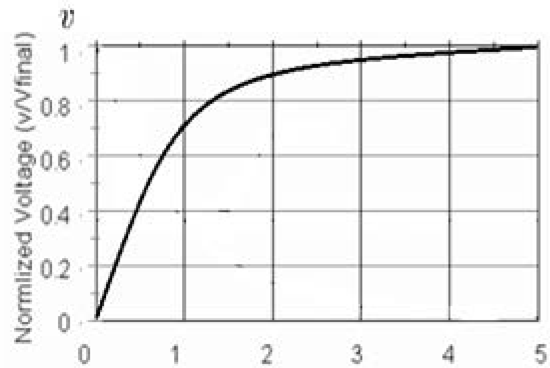


Figure3(System2)

Procedure:

1. Step responses of first order RC and RL circuits are shown in Figure2 and Figure3.
2. Use these responses to calculate time constant of the circuits.
3. Design both circuits using calculated time constant.
4. Measure the actual values of capacitor, inductor and resistor using DMM and LCR meter and record them in Table 1.
5. Construct RC and RL circuits on breadboard as shown in Figure2 and Figure3.Adjust the function generator to produce 2Vpp Square wave.
6. Adjust the frequency in a way that you can measure the time constant.

7. Compare the calculated and measured time constants.
8. Find the transfer functions for both circuits using K and τ.
9. Using the transfer function found in step(8),draw the asymptomatic Bode plots on given semi log paper in Figure 6 and Figure7.



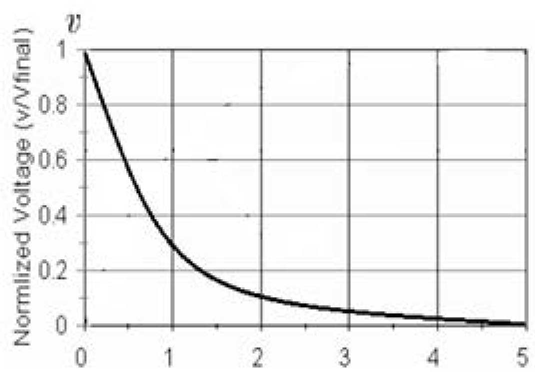


Figure4(Normalized time t/RC)

Figure5(Normalized timetR/L)

Note that the time constant (t= τ = RC) occurs at 0.632Vin

TABLE-1: Calculated Circuit Parameters:

Time Constant	**R=2.2KΩ**	**C=**	τ =
Time Constant	**R=4.7KΩ**	**L=**	τ =
Transfer Function(RC)			
Transfer Function(RL)			

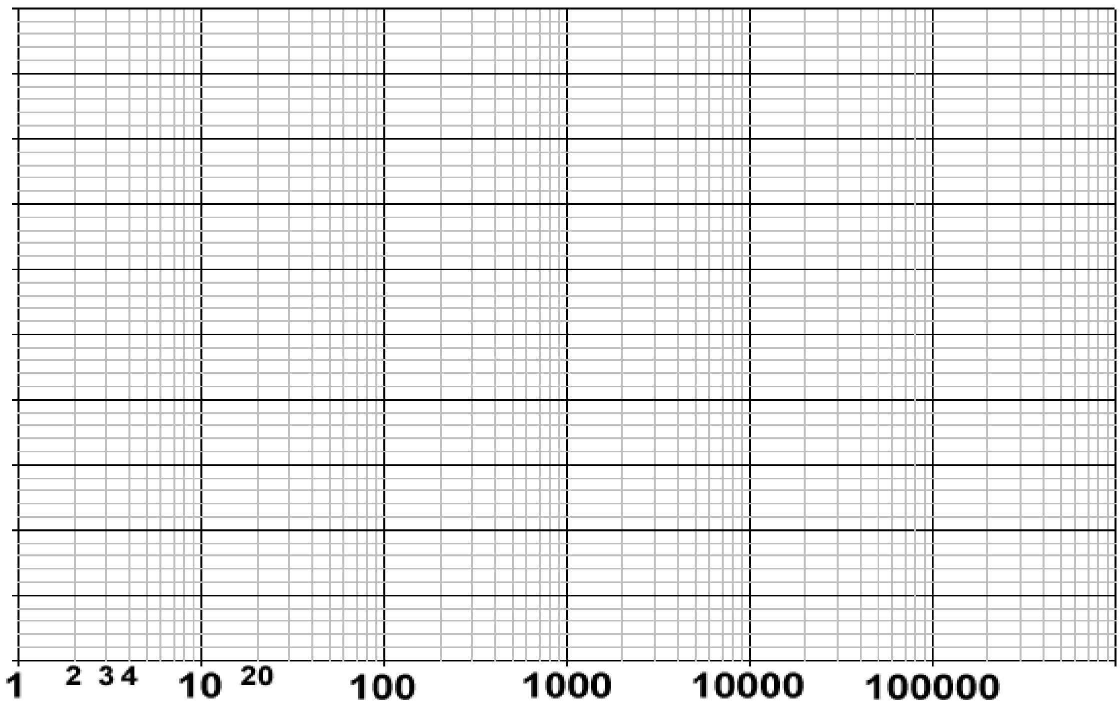


Figure6**(Asymptotic bode plot of RC circuit)**

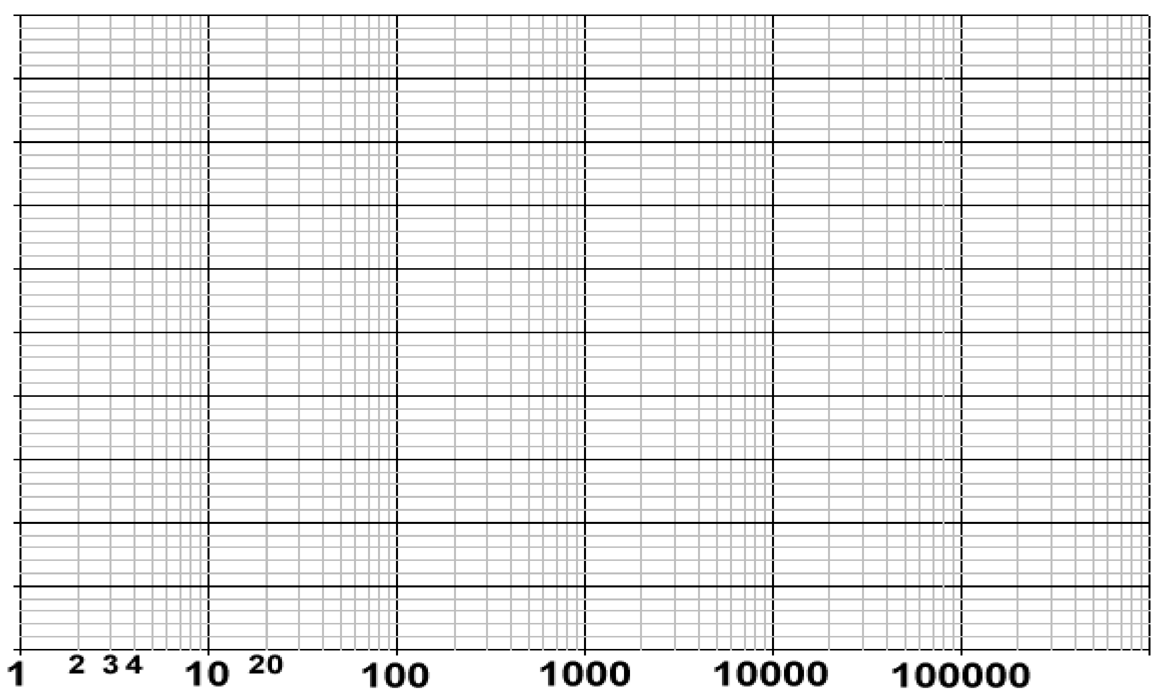


Figure7(Asymptotic bode plot of RL circuit)

Part(b): Frequency Response of First order circuit.

Procedure:

1. Construct RC and RL circuit on breadboard as shown in Figure2 and Figure3.
2. Connect the function generator at input. Adjust the function generator to produce 2Vpp Square wave.
3. Increase the frequency from 500Hz to 10 KHz.
4. Measure all the voltage across capacitor and Inductor.
5. Plot the frequency response of each system on the given semi log paper (Take logarithmic frequency on x-axis and voltage in dBs on y-axis.

TABLE-2: Observations and Calculations

Frequency (Hz)	V_{in}	System1(RC)		System2(RL)	
		V_{out}	V_{out}(dB)	V_{out}	V_{out}(dB)
500					
1000					
2000					
3000					
4000					
5000					
6000					
7000					
8000					
9000					
10,000					

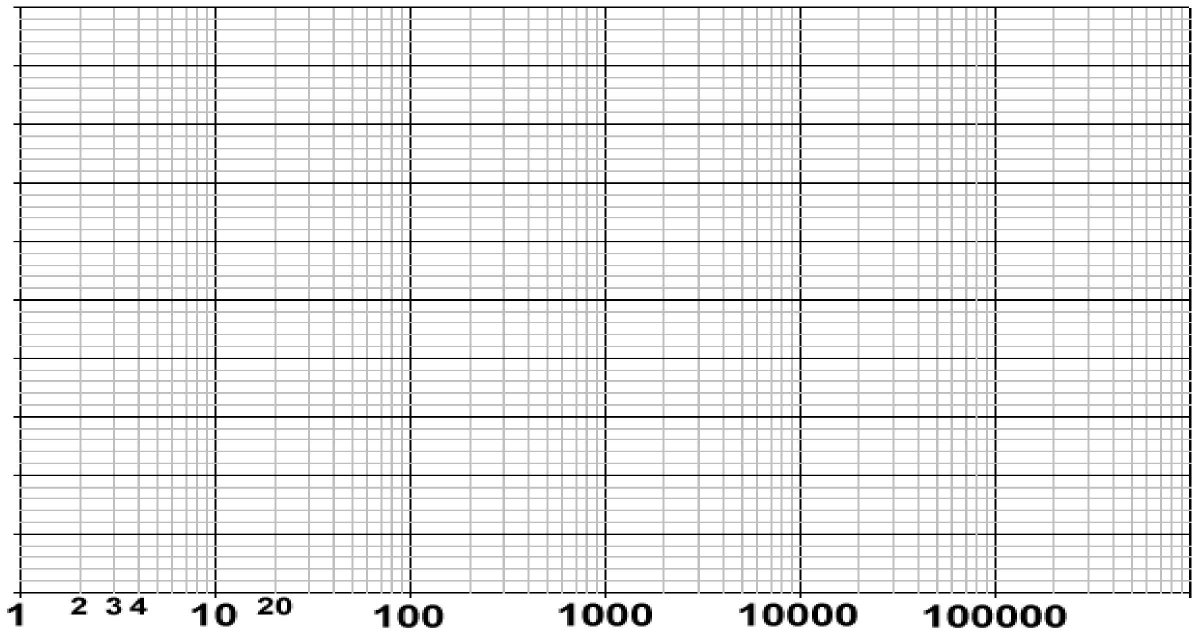


Figure8**(Actual Frequency response of RC circuit**)

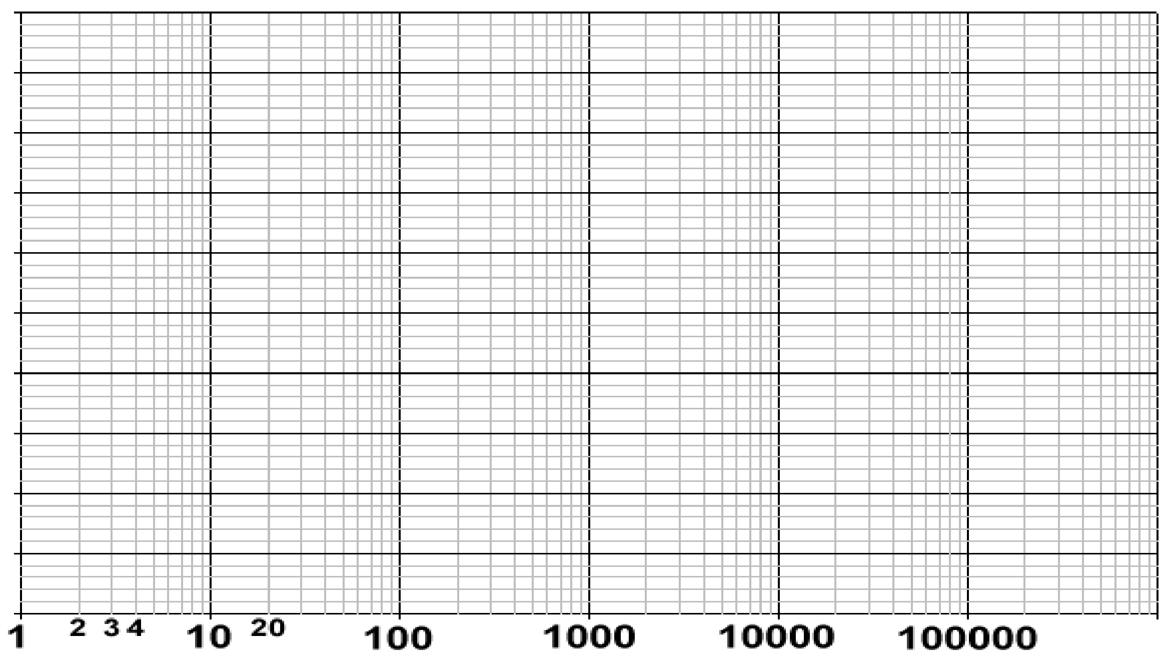


Figure9(Actual Frequency response of RL circuit

Compare asymptotic and acutal frequency response of both circuits?

Post Lab:

Q1. For the first order system given below find

$$G(s) = \frac{3}{s+5}$$

- DC gain K and time constant τ
- Draw the step response of above first order system.

Q2. Comment on the shape of frequency response of each system.Why is it different from the expected shape?

Q3. Construct the bode plot on a semilog paper whose transfer function is given by

$$\frac{10(s+1)}{s+10}$$

EXPERIMENT 2

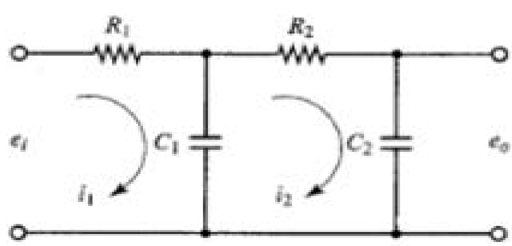
Frequency sweep of second order systems

Objective:

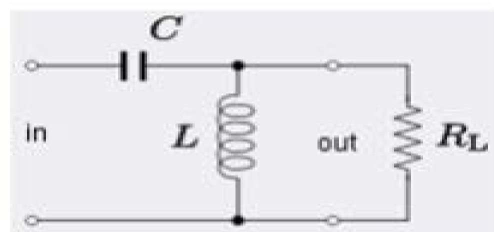
Observe and plot frequency sweep of second order systems.

Equipment Required:

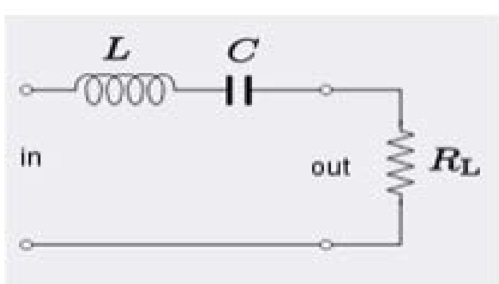
- Oscilloscope with probes
- Function generator with probes
- Digital multi-meter
- Dual Power supply
- Breadboard
- Capacitor 1nF
- Inductor 150mH
- Resistor 1KΩ



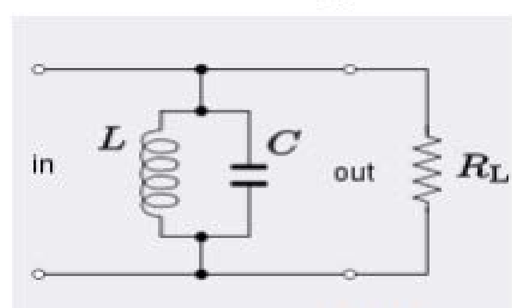
System 1



system 2



system 3



system 4

Figure 1(Second Order Systems)

Procedure:

1. Measure the actual values of capacitor, inductor and resistor using DMM and LCR meter and record them in table 1.
2. Connect the circuit as shown in Figure1.
3. Connect the function generator at input. Adjust the function generator to produce 4Vpp sine wave.
4. Increase the frequency from 100Hz to 40 KHz and complete Table-2.
5. Plot the frequency response of each system on the given semi log paper in Fig2 (Take logarithmic frequency on x-axis and voltage in dBs on y-axis.

DATA FOR EXPERIMENT:

TABLE-1 (MEASURED VALUES)

Resistor	
Capacitor	
Inductor	

TABLE-2(OBSERVATIONS AND CALCULATIONS)

Frequency		**System 1**		**System 2**	
(Hz)	V_{in}	V_{out}	V_{out}**(dB)**	V_{out}	V_{out}**(dB)**
100					
500					
1000					
5000					
10,000					
15,000					
20,000					
25,000					
30,000					
35,000					
40,000					

TABLE-3(DATA FOR EXPERIMENT)

Frequency (Hz)	**V_{in}**	**System 3**		**System 4**	
		V_{out}	**V_{out}(dB)**	**V_{out}**	**V_{out}(dB)**
100					
500					
1000					
5000					
10,000					
15,000					
20,000					
25,000					
30,000					
35,000					
40,000					

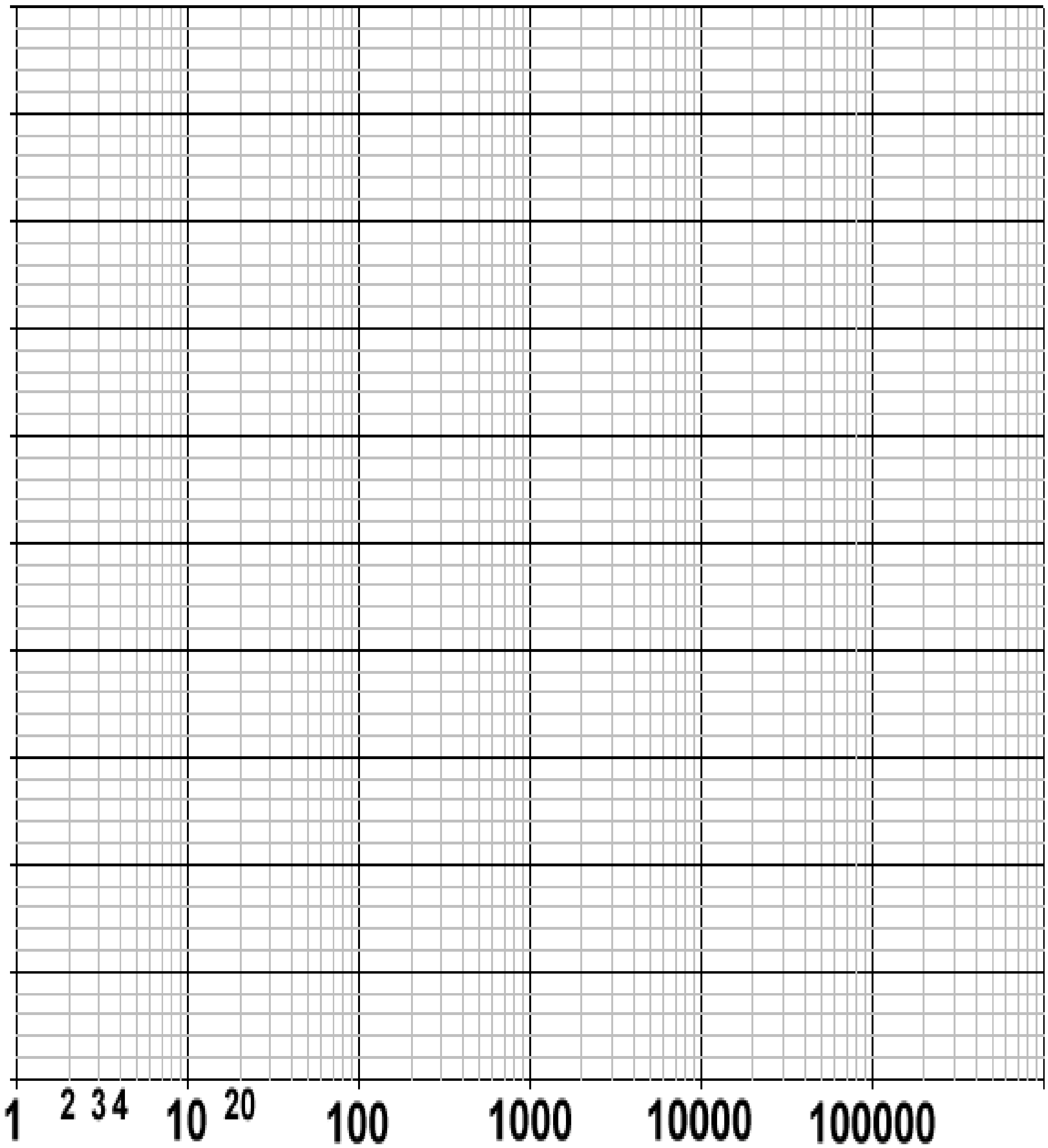


Figure 2 (Frequency response of 2nd order system)

Post Lab:

Q1. Explain the observations made during the performance of the lab experiment.

Q2.Would the frequency sweep of each second order system be altered if the value of inductor or capacitor was changed? If so,why?

EXPERIMENT 3

Design of System's Transfer Function & Response and Introduction to MATLAB SIMULINK

Objective:

1. To understand the MATLAB functions used to define the transfer function and response of a system and to solve complicated polynomials.
2. To find the inverse Laplace transform and to compute partial fraction expansion of the ratio of two polynomials.
3. To understand MATLABSIMULINK and implement system's transfer function using it.
4. To solve the system equations and obtain the response of the system for different inputs.

3.1 Transfer Function with MATLAB

Use the following MATLAB commands in all the in-Lab tasks. Use MATLAB 'help' to find the purpose of these commands

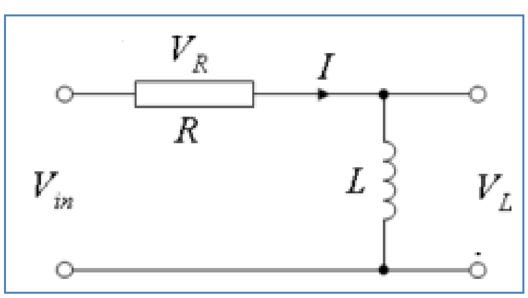
a. sys=tf(num, den)
b. bode(sys)
c. step(sys)
d. impulse(sys)

Tasks:

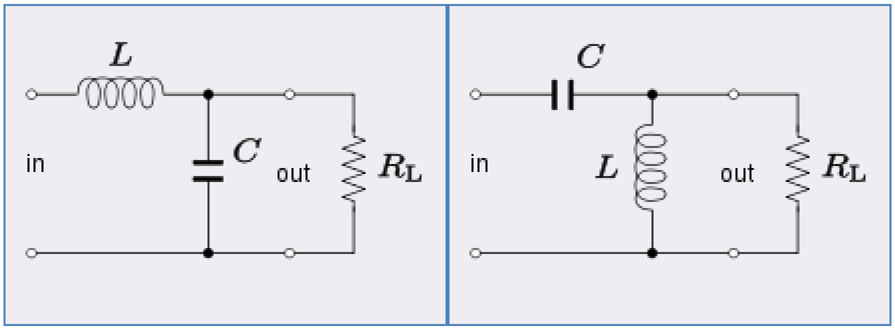
1. For the following circuits shown in Figure 3.1, Find

i. Transfer Function
ii. Identify whether it is low pass, high pass, band pass or band reject filter
iii. Step response
iv. Impulse response

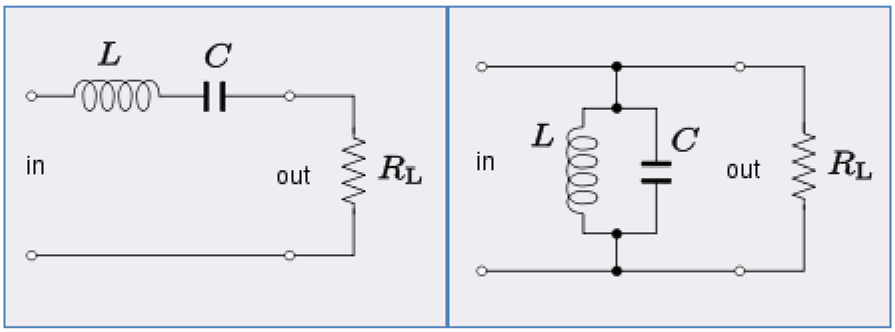
Use $R = 3,\ L = 4\ and\ C = 2$



(a)



(b) (c)



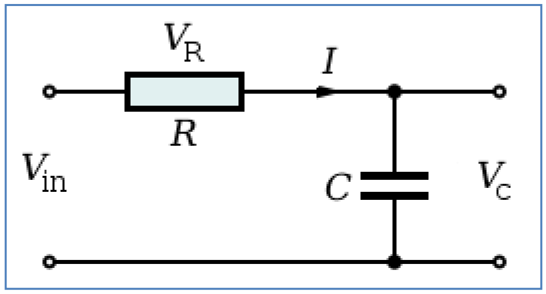
(d) (e)

Figure 3.1

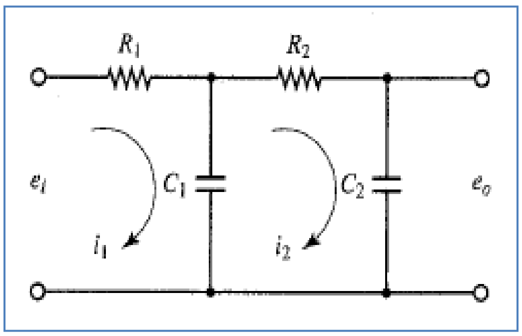
2. For the following circuits shown in Figure 3.2, Find

i. Transfer Function
ii. Identify whether it is low pass, high pass, band pass or band reject filter
iii. Step response
iv. Impulse response

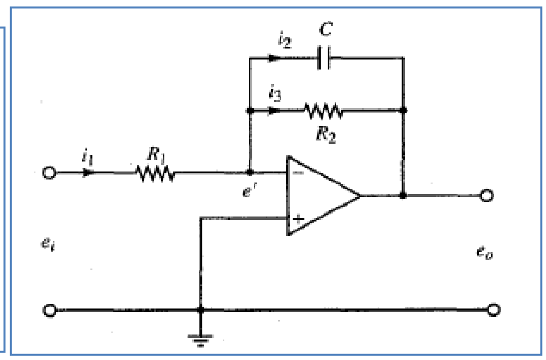
Use$R_L = 3$; $L = 4$; $C = 2$; $R_1 = R_2 = 4$; $R_3 = 2$; $R_4 = 4$



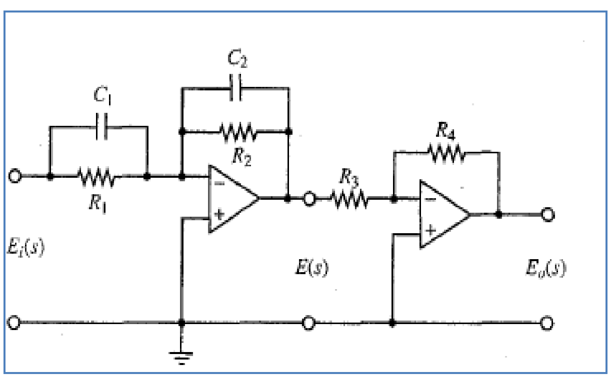
(a)



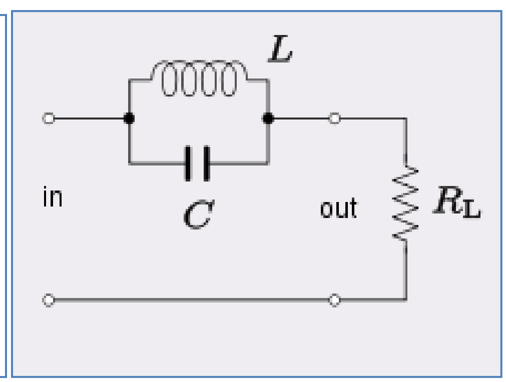
(b)



(c)



(d)



(e)

Figure 3.2

3.2Partial Fraction Expansion with MATLAB

When trying to find the inverse Laplace transfer (or inverse z transform), it is helpful to be able to break a complicated ratio of two polynomials into forms that exist on the Laplace Transform (or z transform) table. The “residue” function of MATLAB can be used to compute the partial fraction expansion (PFE) of the ratio of two polynomials. It takes numerator and denominator vectors as input which are coefficients of the numerator and denominator polynomials respectively. The “tf2zp” function returns poles, zeros and K of a transfer function. The “zp2tf” function returns the numerator and denominator coefficients when zeros, poles and K are sent as input parameters.

a. [r,p,k] = residue(num,den)
b. [z,p,K]=tf2zp(num,den)
c. [num,den]= zp2tf(z,p,K)

Exercise 1:

Obtain the inverse Laplace transform of the following F(s). [Use MATLAB to find the partial fraction expansion of F(s)]. Write the inverse Laplace transform in the text box below

$$F(s) = \frac{s^5 + 8s^4 + 23s^3 + 35s^2 + 28s + 3}{s^3 + 6s^2 + 8s}$$

Exercise 2:

Given the zero(s), pole(s), and gain K of B(s)/A(s), obtain the function B(s)/A(s) using MATLAB. Consider the three cases below. Write the transfer function of each in the text box below:

- **There is no zero. Poles are at *-1+2j* and *-1-2j, K=10***
- **A zero is at 0. Poles are at *-1+2j* and *-1-2j, K=10***
- **A zero is at -1. Poles are at *-2,-4and -8. K=12.***

Exercise 3:

A function B(s)/A(s) consists of the following zeros, poles, and gain K:

- **Zeros at s=-1, s=-2**
- **Poles at s=0, s=-4, s=-6**
- **Gain K=5**

Obtain the expression for B(s)/A(s) = num /den with MATLAB and write it in the space below:

Exercise4:

Obtain the partial fraction expansion of the following function with MATLAB:

$$F(s) = \frac{10(s+2)(s+4)}{(s+1)(s+3)(s+5)^2}$$

Then, obtain the inverse Laplace transform of F(s). Write the inverse Laplace transform in the space below:

Exercise5:

Consider the following function F(s):

$$F(s) = \frac{s^4+5s^3+6s^2+9s+30}{s^4+6s^3+21s^2+46s+30}$$

Using MATLAB, obtain the partial fraction expansion of F(s). Then, obtain the inverse Laplace transform of F(s) and write it in the box below:

<u>Getting started with MATLABSIMULINK</u>

3.3 SIMULINK Tutorial

SIMULINK is the Graphical User Interface (GUI) for MATLAB. This section presents a brief tutorial on how to use SIMULINK to create an open-loop block diagram.

1. Start MATLAB and type “simulink” (all lower case) in the command window

2. If installed, the SIMULINK Library Browser will soon pop up as shown in Figure 3.3.

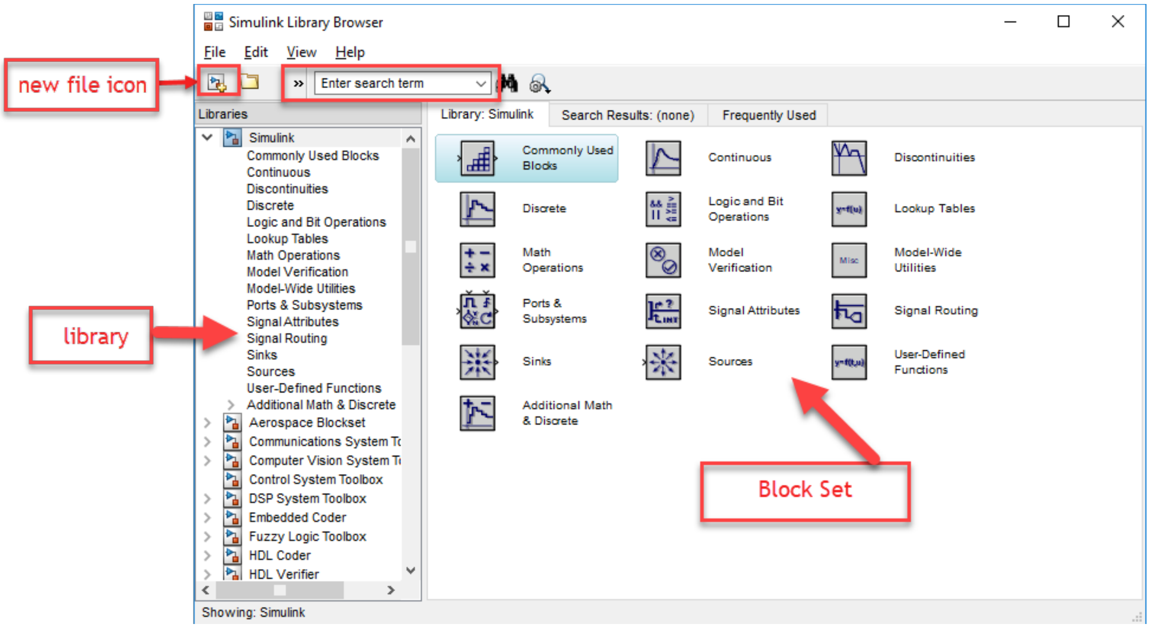


Figure 3.3

3. Click on the "new file icon" as shown in Figure 3.3, identical to a MS Word new file icon. A new window will appear as show in Figure 3.4. That is your space to work in. After creating a model it can be saved (using the save icon).

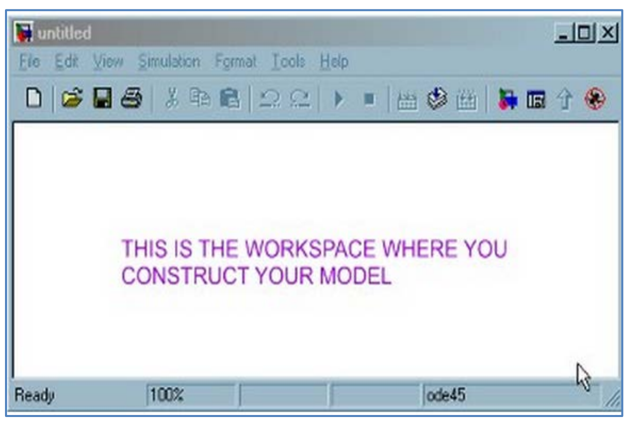


Figure 3.4

4. To build simulation models, you will be creating block diagrams. In general, all blocks are double-clickable to change the values within. You can connect the ports on each block via arrows easily by clicking and dragging with the mouse. You can also double-click any arrow (these are the controls variables) to label what it is. Same with all block labels (SIMULINK will give a default name that you can change).

5.SIMULINK uses EE lingo. Sources are inputs and sinks are outputs. If you click around in the SIMULINK Library Browser, you will see the possible sources, blocks, and sinks you have at your disposal.

Example:

Now let us create a simple one-block transfer function and simulate it subject to a unit step input.

The given open-loop transfer function is $G(s) = \frac{1}{s^2+2s+8}$

a. Click the new icon in the SIMULINK Library Browser to get a window to work in (untitled with the SIMULINK logo).

b. Double-click the "Continuous" button in the SIMULINK Library Browser to see what blocks are provided for continuous control systems. Grab and slide the "Transfer Fcn" block to your workspace(Refer to Figure 3.5).

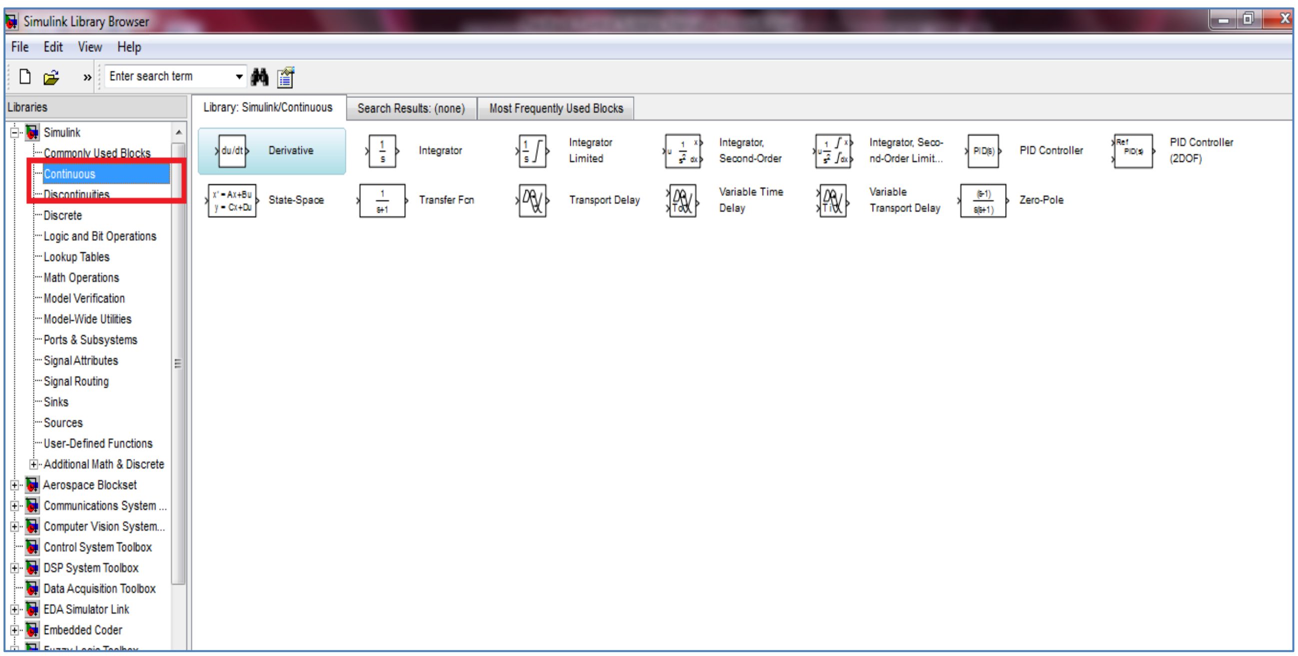


Figure 3.5

c. Double-click the block in your workspace and enter 1 in Numerator coefficients and 1 2 8 in Denominator coefficients and close by clicking OK. SIMULINK will update the transfer function in the block, both mathematically and visually.

d. Go ahead and save your model as name.mdl (whatever name you want, as long as it is not a reserved MATLAB word).

e. Click the Sources tab in the SIMULINK Library Browser to see what source blocks are provided. You will find a Step, Ramp, Since Wave, etc. Grab and slide the Step block to your workspace as shown in Figure 3.6.

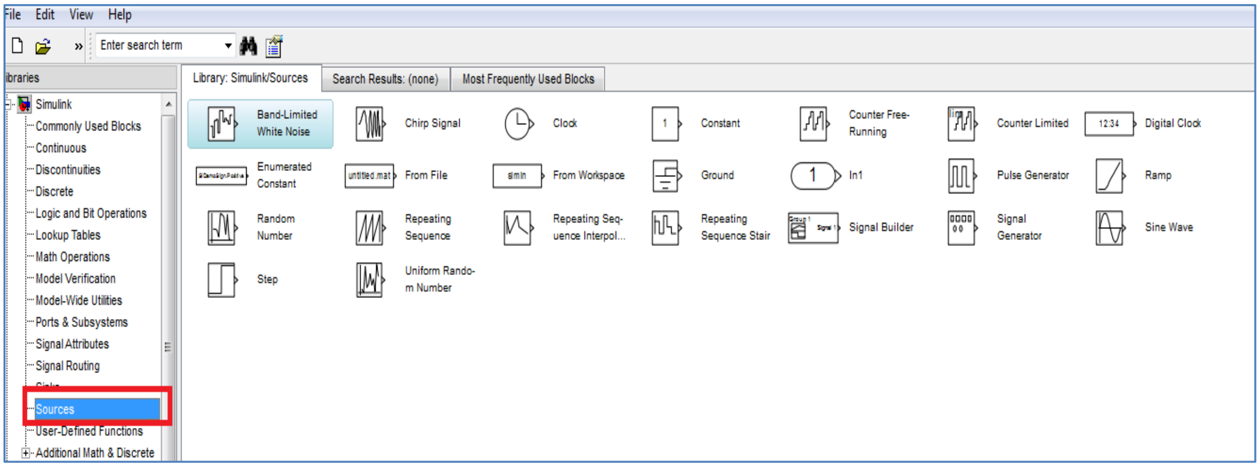


Figure 3.6

Double-click the Step block in your workspace and ensure 1 is entered as the final value (for a unit step) and that 0 is the initial value. Close by clicking OK.

f. Draw arrows

g. Click the Sinks tab in the SIMULINK Library Browser to see what sink blocks are provided. Grab and slide the Scope block to your workspace as shown in Figure 3.7.

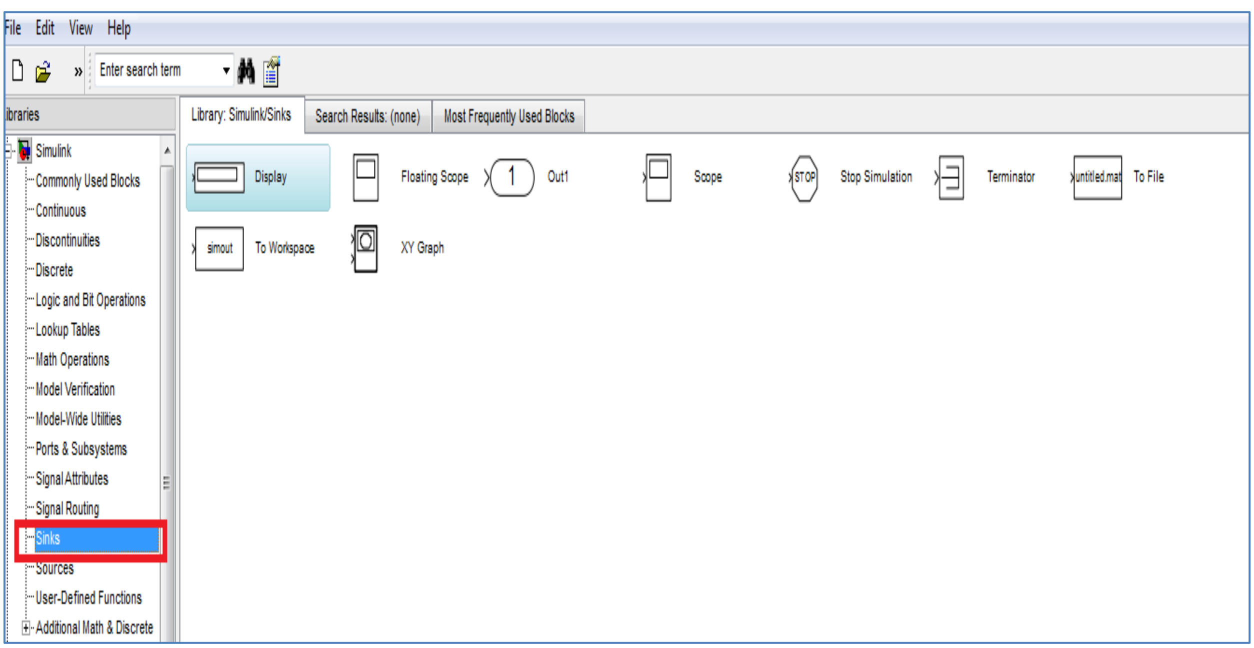


Figure 3.7

h. Draw an arrow from the "Transfer Fcn" block to the Scope block by using the mouse, the same method as before.

i. To run the model (solve the associated differential equation numerically and plot the output results vs. time automatically), simply push play (the solid black triangle button in your workspace window).

j. After it runs, double-click on your Scope to display the results.
Your final model will look like Figure 3.8:

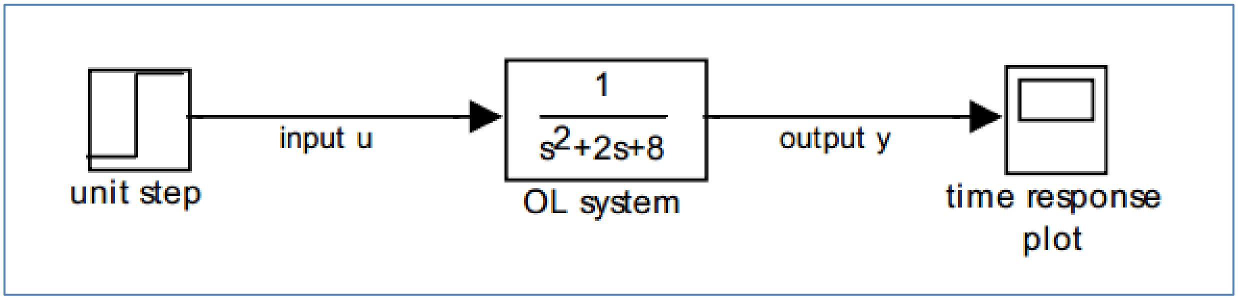


Figure 3.8

Modeling an impulse in SIMULINK:

Square pulse can be used to generate an approximation of a basic impulse. It simply requires two step blocks to simulate an impulse in SIMULINK. Connect the blocks as shown in Figure 3.9.

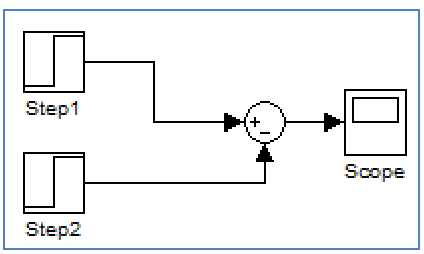


Figure 3.9 Step block impulse model

Now, set the step block parameters to the values shown in table3.1.

Block	Step Time	Initial Value	Final Value
Step1	1	0	10
Step2	1.01	0	10

Table 3.1

After running the model, the results should appear as in Figure 3.10

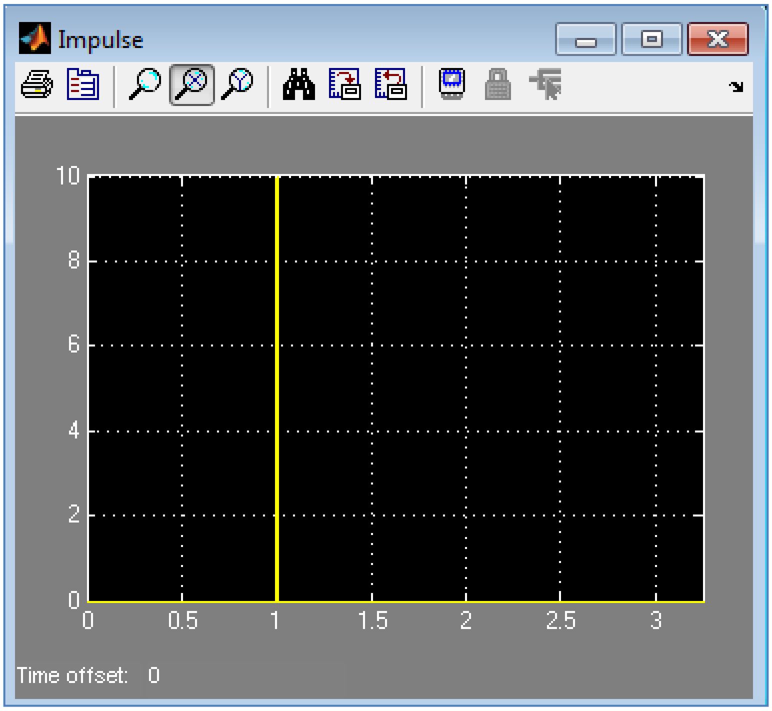


Figure 3.10: Step Block impulse

This approximation of impulse signal can be used to find the impulse response of any dynamic system.

Exercise1:

Generate the following MATLABSIMULINK model in Figure 3.11 and simulate its step response.

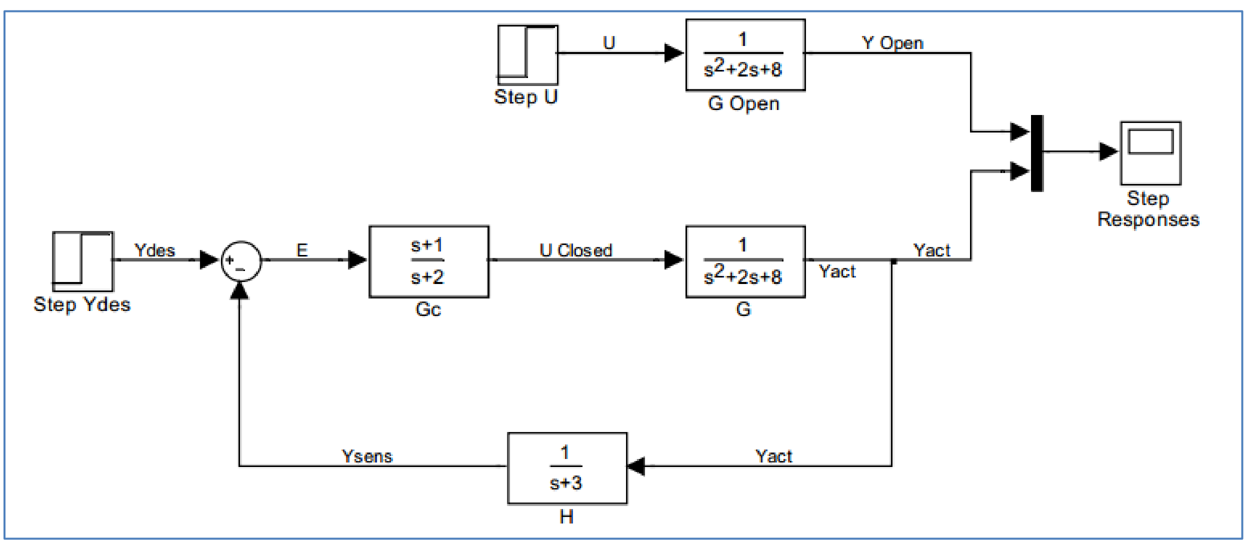


Figure 3.11

Exercise 2:

a. Obtain the unit impulse response of the following system using SIMULINK.

$$\frac{B(s)}{A(s)} = \frac{1}{s^2 + 0.2s + 1}$$

b. Obtain the unit step response of the following system using SIMULINK.

$$\frac{B(s)}{A(s)} = \frac{s}{s^2 + 0.2s + 1}$$

Explain why the results in ‘a’ and ‘b’ are same.

Post Lab

Question 1:

Solve the following differential equation using MATLAB:

$$\ddot{x} + 2\dot{x} + 10x = e^{-t}, x(0) = 0, \dot{x}(0) = 0$$

The function e^{-t} is given at t=0 when the system is at rest.

Question 2: Find the transfer function of following circuit shown in Figure 3.12:

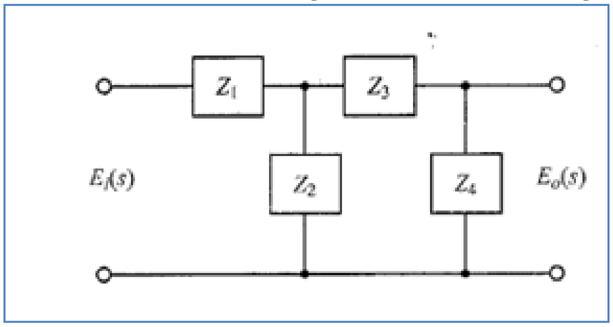


Figure 3.12

Question 3:

a) How can LTI filters be uniquely identified by their impulse response?

b) How can you identify the order of a systemfrom its differential equation?

c) How can Laplace transform offer a convenient method for the solution of linear, time-invariant differential equations?

<u>Question 4:</u>

Create a SIMULINK model with a first order system, with gain, K =1, and time constant, T = 0.1 sec. Simulate a square wave input with unit amplitude and frequency of 0.3 Hz. The sample time is 0.001 sec. View the reference position, xr(t), input, u(t), and actual position, x(t), through a scope, as in Figure 3.13 below. Experiment with different values of K_p and observe how the system response changes. Plot the results.

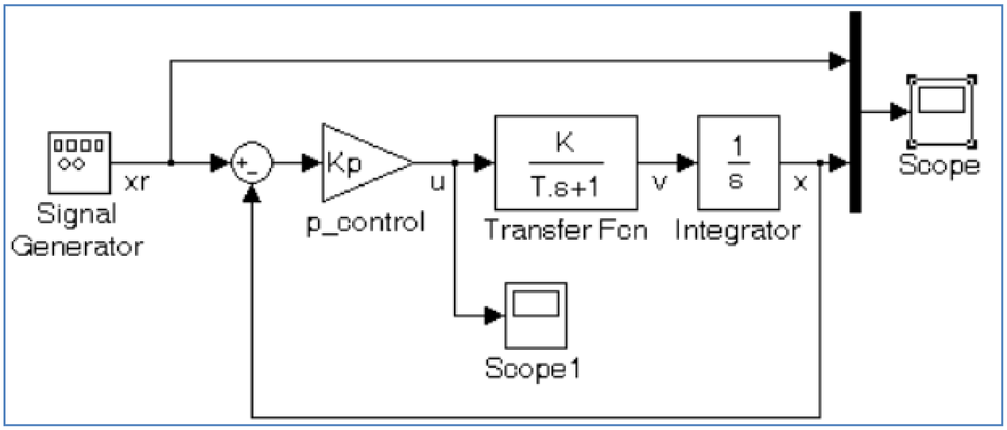


Figure 3.13

EXPERIMENT4

Mathematical Modeling of Physical Systems

Objective:

1. To understand the role of mathematical models of physical systems in design and analysis of control systems.
2. To learn MATLAB functions in solving and simulating such models.

4.1 Mechanical System: Mass-Spring System Model

Consider the following Mass-Spring system shown in the Figure4.1 below.

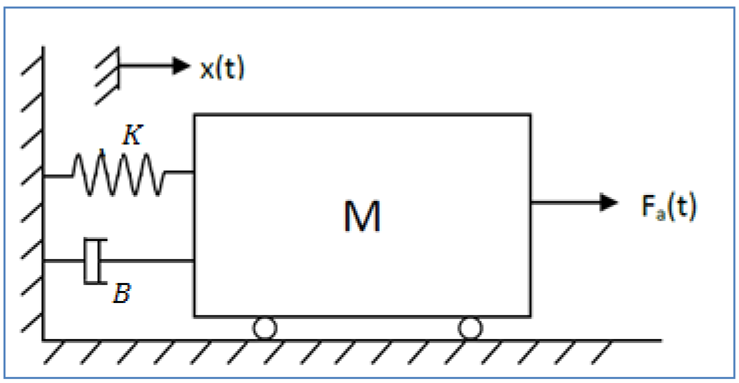


Figure 4.1

K is spring constant.

B is friction coefficient

x(t) is the displacement

$F_a(t)$ is the applied force

Exercise1:

a) Derive the second order differential equation of the mass-spring system shown in Figure 4.1 and write it down in the space below:

b) Write the transfer function of the system:

X(s)/F(s)=

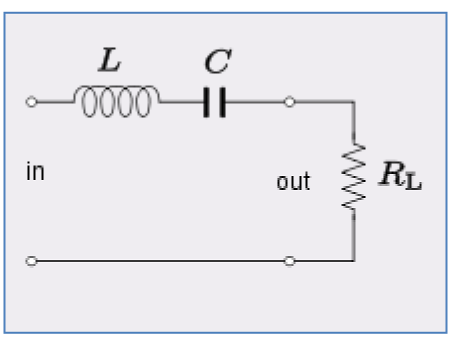
c) Write the state space equation of the above system in the space below:

d) Construct a SIMULINK diagram to calculate the response of the Mass-Spring system. The input force increases from 0 to 8 N at t = 1 s. The parameter values are M = 2 kg, K= 16 N/m, and B =4 N.s/m (Draw a block diagram to represent this equation and draw the corresponding SIMULINK diagram before implementing it on SIMULINK).

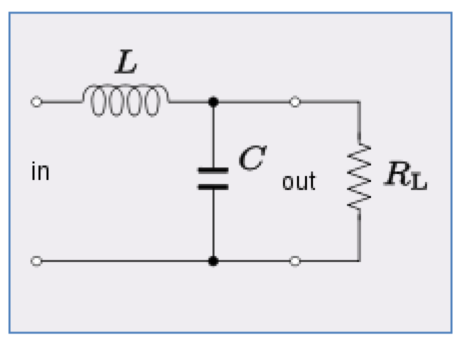
4.2 Electrical System: RLC circuit

Exercise 2:

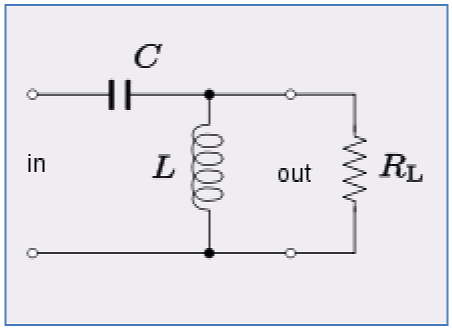
Consider the electrical circuits shown in the Figure 4.2 below.



(a)



(b)



(c)

Figure 4.2

a) Derive the differential equation of the above systems and write it down in the space below, also state its order:

b) Write the transfer function of the systems in the space below:

c) Write the state space equation of the above system in the space below:

d) Construct a SIMULINK diagram to calculate the response of the above systems. Use $R = 3$, $L = 4 \; and \; C = 2$ (Draw a block diagram to represent the equations and draw the corresponding SIMULINK diagram before implementing it on SIMULINK).

Post Lab

A permanent magnet dc motor is a mechanism which converts electrical power to mechanical power via magnetic coupling. The electrical power is provided by a voltage source, while the mechanical power is provided by a spinning rotor. A very basic dc motor is constructed of two main components: the rotor or armature and the stator. The armature rotates within the framework of the stationary stator. A simple illustration of a dc motor is shown in Figure 4.3 below:

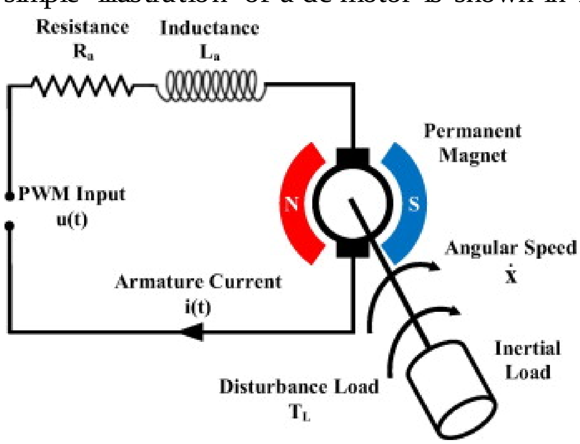


Figure 4.3

The equivalent electrical circuit of a dc motor is illustrated in Figure4.4 below:

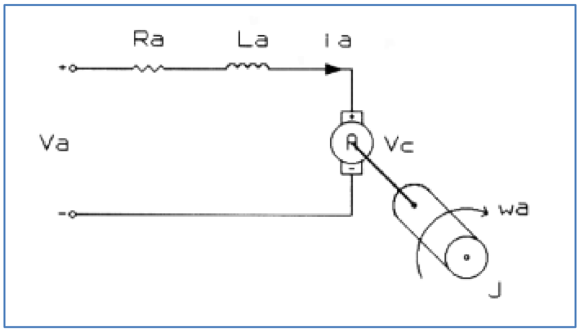


Figure 4.4

The voltage across the inductor is proportional to the change of current through the coil with respect to time.

a) Find the differential equation of the motor, transfer function and state space equations.

b)Make the block diagram of the transfer function and implement it inMATLABSIMULINK. You can reduce the block diagram according to your understanding and you are required to explain each step of reduction in detail.

c)Write detailed analysis on the step response of the DC motor. State the assumptions in your report.

EXPERIMENT 5

Block Diagrams Reduction using MATLAB

Objective:

1. To obtain transfer functions of complex block diagrams through MATLAB.
2. To plot the responses of systems

5.1 Transfer Functions of Block Diagrams and Step Response

In control systems analysis, we frequently need to simplify a network of interconnected transfer functions into a single transfer function which is then used in subsequent calculations for analysis purposes. There are three different types of connections between transfer function that are usually encountered in practice: cascade-connected, parallel-connected and feedback-connected (closed-loop) transfer functions. MATALB has convenient commands to obtain these transfer functions. To obtain the transfer functions of the cascaded, parallel, feedback and unity feedback systems, the following commands are used, respectively:

[num, den] = series (num1, den1, num2, den2)

[num, den] = parallel (num1, den1, num2, den2)

[num, den] = feedback (num1, den1, num2, den2)

[num, den] = cloop (num1, den1, -1)

Exercise1:

Obtain Y(s)/X(s) = num /den for each of the arrangement of $G_1(s)$ and $G_2(s)$ as shown in Figure 5.1 and write the transfer function in front of each arrangement

1. $\boldsymbol{G_1(s) = \frac{10}{s^2+2s+10}}$

2. $G_2(s) = \dfrac{5}{s+5}$

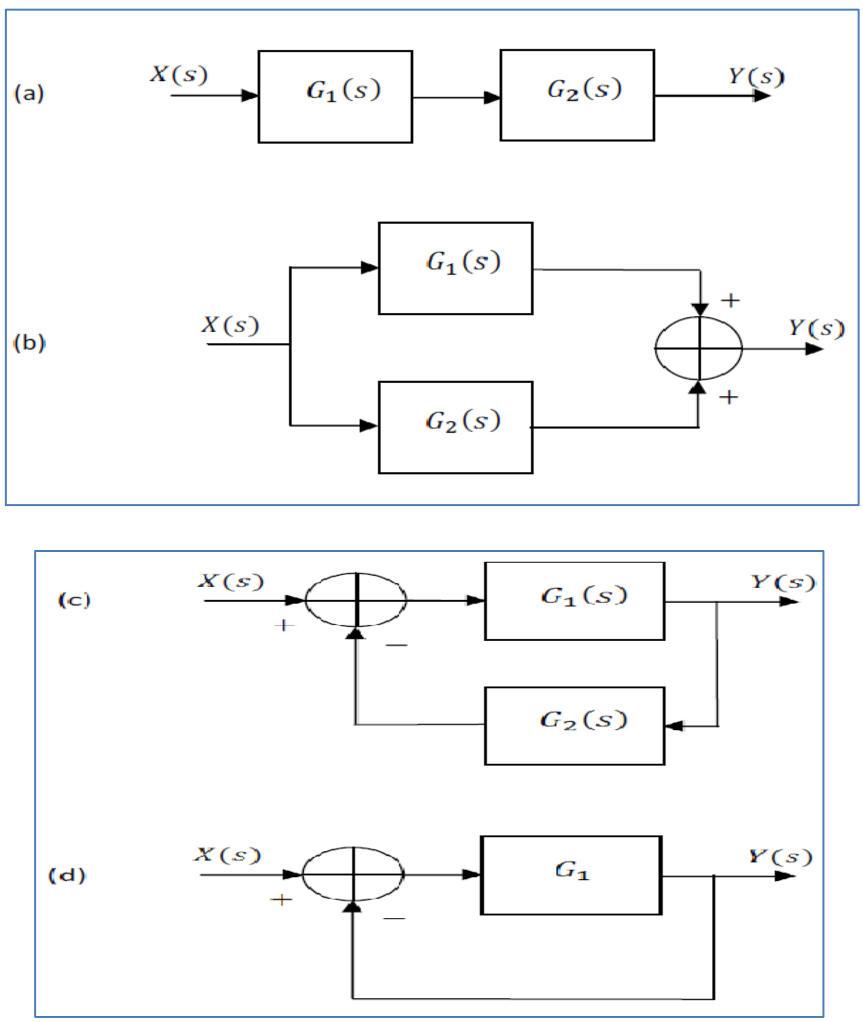


Figure 5.1

Exercise 2:

Evaluate the transfer function of the feedback system shown in the Figure 5.2 using MATLAB. G1(s) = 4, G2(s) = 1/(s+2), H(s) = 5s.Write the code and the transfer function below

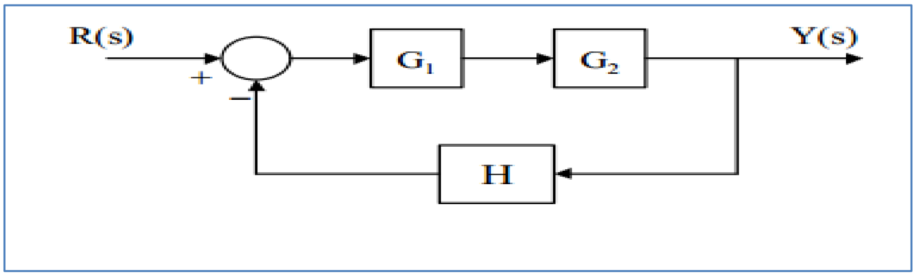


Figure 5.2

Transfer function:

Exercise 3:

Determine the transfer function of the system shown in Figure 5.3. Check your answer with MATLAB

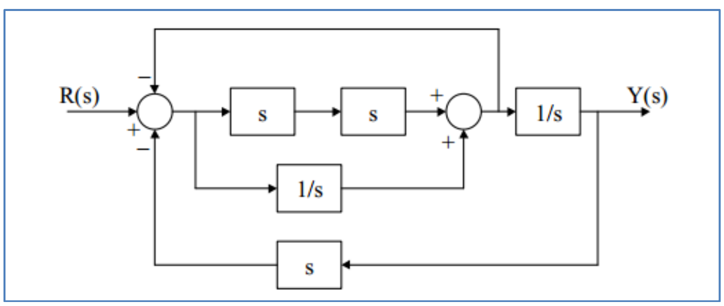


Figure 5.3

Transfer function:

5.2 Transformation of Mathematical Models with MATLAB

[A,B,C,D]= tf2ss(num, den)

[num,den]= ss2tf (A,B,C,D)

[z,p,k]=ss2zp (A,B,C,D,i)

[num, den] = tfdata (T,'v')

Exercise 4:

Transform the transfer function into the state space representation using MATLAB

(a) $Y(s)/U(s) = \frac{s}{(s+10)(s^2+4s+16)}$

(b) $Y(s)/U(s) = \frac{s^2+7s+2}{s^3+9s^2+26s+24}$

Write the state space representation of each transfer function in the box below.

Exercise 5:

Obtain the transfer function of the state space equations using MATLAB

(a)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 8 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

(b)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Exercise 6:

Obtain the closed-loop transfer function and the function to obtain the closed-loop state space model of the system shown below, using MATLAB.

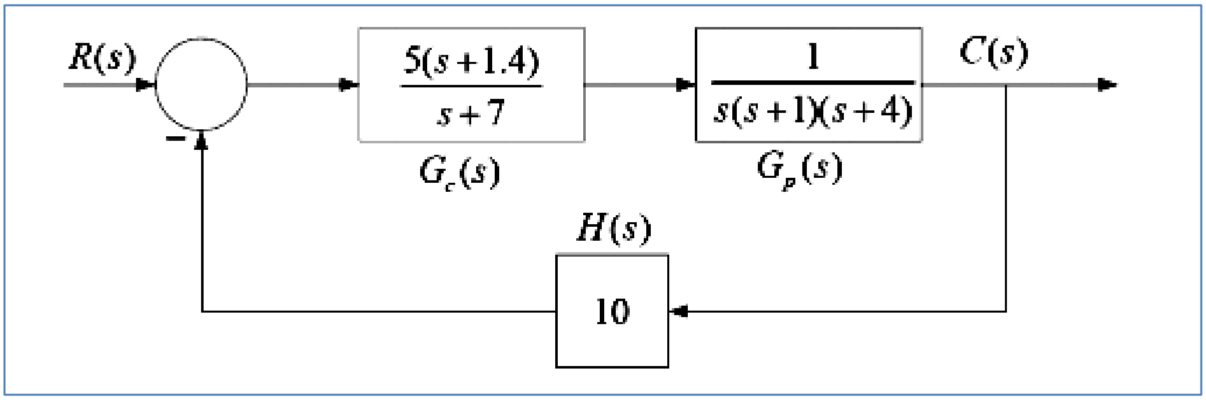


Figure 5.4

Transfer function:

State-space Equation:

Post Lab

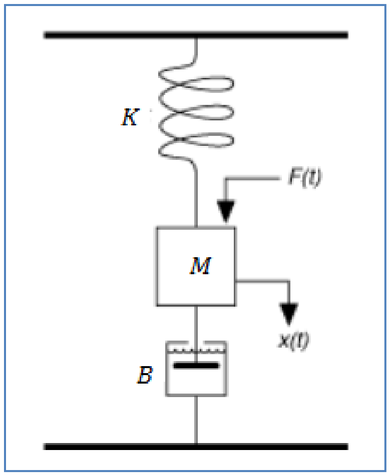


Figure 5.5

Given a mass-spring damper system

B=damping constant

M=mass

K=spring constant

F=u=force

The state space model for this system is shown below

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}0 & 1\\ -\frac{k}{m} & -\frac{c}{m}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}0\\ \frac{1}{m}\end{bmatrix}u$$

$$y=\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

a) Define the state-space model above using the ss function in MATLAB.

Set c=1, m=1, k=50.

b) Use MATLAB to simulate the result. Explain each step in detail.

c) Find the transfer function from the state-space model.

d) Find the responses of the system using SIMULINK and insert the screen shots.

EXPERIMENT6

Second Order System Analysis using MATLAB

Objective:

1. To implement the systems in MATLAB
2. To understand the system transient and steady-state responses

The steady-state response means the manner in which the system output behaves as t approaches infinity. The transient response means that which goes from the initial state to the final state.

6.1 Transient Response

Exercise1: Write MATLAB program to obtain the unit-step response curves for the following systems:

(a).

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 8 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

b).

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Exercise 2: For ζ=0.4 and $\omega_n = 5\ rad/sec$. Obtain the standard second order system'stransfer function and write it in the box below:

Exercise 3: For the system found in Question 2, write a MATLAB program to find the rise time, peak time, maximum overshoot and the settling time. Use **ltiview()**command.

Peak Time:
Maximum Overshoot:
Rise Time:
Settling Time:

6.2 Steady State Response

Exercise 4: Obtain the partial fraction expansion of C(s) using MATLAB when R(s) is a unit step function. Show the system response c(t) curve using ltiview() command.Also write down the time response c(t) in the space below:

$$\frac{C(s)}{R(s)} = \frac{3s^3 + 25s^2 + 72s + 80}{s^4 + 8s^3 + 40s^2 + 96s + 80}$$

Exercise 5: When the closed-loop system involves a numerator dynamics, the unit-step response curve may exhibit a large overshoot. Comment on this statement after obtaining the unit-step response and the unit-ramp response of the following system with MATLAB.

$$\frac{C(s)}{R(s)} = \frac{10s + 4}{s^2 + 4s + 4}$$

Exercise 6: Using MATLAB, obtain the unit-ramp response of the closed-loop control system whose closed-loop transfer function is given below. Also, obtain the response of this system when the input is given by: $\boldsymbol{r = e^{-0.5t}}$

$$C(s)/R(s) = \frac{s + 10}{s^3 + 6s^2 + 9s + 10}$$

Exercise 7: Consider the system show in Figure 4.1 below. Determine the value of k such that damping ratio is 0.5. Using MATLAB obtain the rise time, peak time, maximum overshoot, and settling time in the unit-step response and fill the following table.

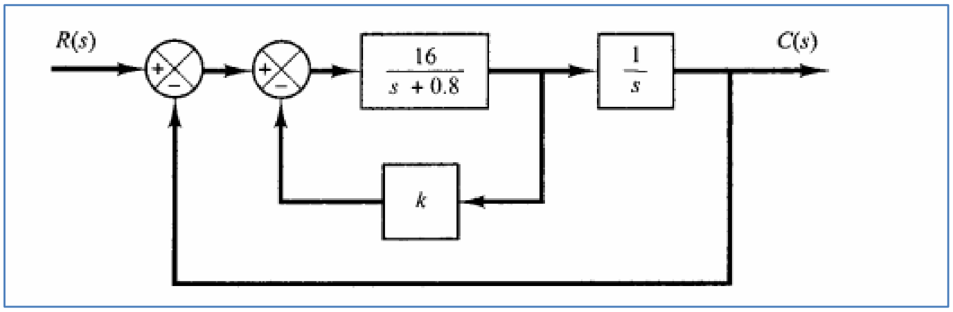


Figure 6.1

Rise Time:	
Peak Time:	
Maximum overshoot	
Settling Time	
Value of k	

Post Lab

Q1: Using MATLAB, obtain the unit-step response, unit-ramp response, and unit impulse response of the system defined below. Where R(s) and C(s) are LAPLACE transform of the input r (t) and output c(t) respectively.

$$C(s)/R(s) = \frac{10}{s^2 + 2s + 10}$$

Q2: Consider the differential equation system given by:

$$\ddot{y} + 3\dot{y} + 2y = 0, \qquad y(0) = 0.1, \qquad \dot{y}(0) = 0.05$$

Using **syms** and **dsolve** commands, obtain the response y(t) subject to the given initial condition. Use MATLAB help to understand these commands.

EXPERIMENT 7

Modeling and System Identification of a DC Motor

Objective:

- Study the characteristics of step response of first order servo system
- Measure the steady state gain K and time constant τ
- Familiarize with QNET DC motor trainer

7.1 DC Motor Modeling

Modeling the dynamical properties of a system is an important step in analysis and design of control systems. In this experiment, you will study the steady-state and dynamic characteristics of a DC motor. Here a motor-generator combination serves as a first-order linear system. It is important from a practical point of view as many real-world systems can be modeled in this manner. The DC motor considered in this experiment is just one such example.

The electrical equations describing the open-loop response of the DC motor are

$$V_m(t) - R_m I_m(t) - E_{emf}(t) = 0 \qquad [1]$$

and

$$E_{emf}(t) = K_m \omega_m(t) \qquad [2]$$

The mechanical equations describing the torque of the motor are

$$T_m(t) = J_{eq}\left[\frac{d}{dt}\omega_m(t)\right] \qquad [3]$$

and

$$T_m(t) = K_t I_m(t) \qquad [4]$$

Where T_m, J_{eq}, ω_m, K_t, K_m, and I_m are described in Table7.1. This model does not take into account friction or damping.

Symbol	***Description***	***Units***
V_m	Motor terminal voltage	V
R_m	Motor terminal resistance	Ω
I_m	Motor armature current	A
K_t	Motor torque constant	$\mathrm{N.m/A}$
K_m	Motor back-electromotive force constant	$\mathrm{V/(rad/s)}$
ω_m	Motor shaft angular velocity	$\mathrm{rad/s}$
T_m	Torque produced by the motor	$\mathrm{N.m}$
J_{eq}	Motor armature moment of inertia and load moment of inertia	$\mathrm{kg.m^2}$

Table 7.1:DC Motor Model Parameter

The mathematical model of DC motor, $G(s)$,can derived by solving equations [1],[2],[3], and [4] . The following first order transfer function $G(s)$, represents the DC motor angular speed ω_m to the input voltage V_m .

$$G(s) = \frac{\omega_m(s)}{V_m(s)} = \frac{K}{\tau s+1} \qquad [5]$$

Where τ is the time constant of the system, and K is the steady state gain(also known as DC gain). Theseare the two parameters that characterize the first order system.The block diagram is shown in Figure7.1

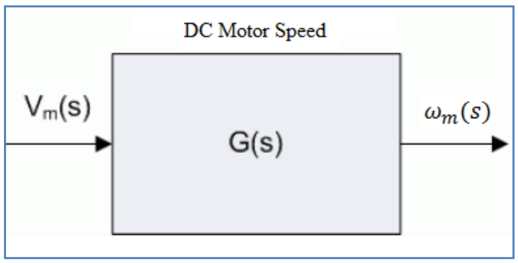


Figure7.1: Open-loop Transfer Function Diagram of Motor Speed

7.2 Experimental Determination of System Dynamics:

The main objective of this experiment is to determine the first order system parameters from its step response.Determining system parameters from measured response data is known as system identification.Bumptest method can be used to determine the system parameters from measured step response.

7.3Bumptest Method

The bumptestis a simple test based on the step response of a stable system. A step input is given to the system and its response is recorded. As an example, consider the system given by the following transfer function:

$$\frac{Y(s)}{U(s)} = \frac{K}{\tau s+1} \qquad [6]$$

The step response shown in Figure 7.2 is generated using the transfer function$\frac{Y(s)}{U(s)}$.

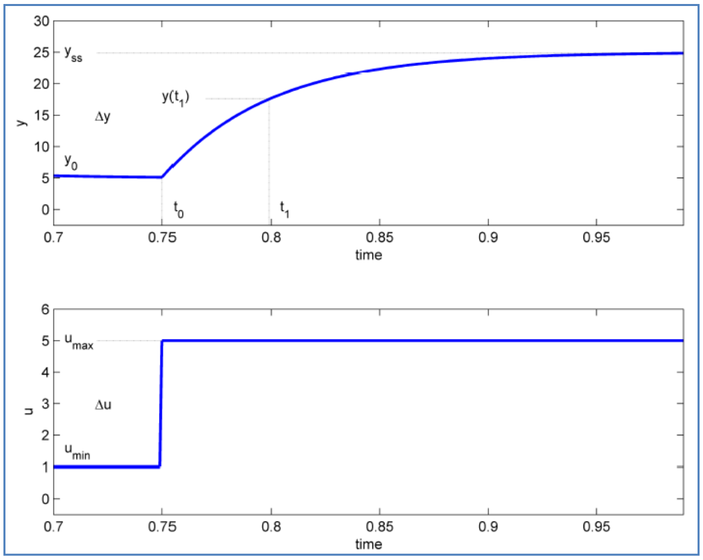


Figure 7.2: Input and output signal used in the bumptest method

The step input begins at time t_0. The input signal has a minimum value u_{min} and a maximum value u_{max}. The resulting output signal is initially at y_0. Once the step is applied, the output tries to follow it and eventually settles at its steady-state value y_{ss}. From the output and input signals, steady-state gain is

$$K = \frac{\Delta y}{\Delta u} \quad [7]$$

Where $\Delta y = y_{ss} - y_0$ and $\Delta u = u_{max} - u_{min}$. In order to find the model time constant, τ, we can first calculate where the output is supposed to be at the time constant form:

$$y(t_1) = 0.632 y_{ss} + y_0 \quad [8]$$

Then, we can read the time t_1 that corresponds to $y(t_1)$ from the response data in Figure a. From the figure we can see that the time t_1 is equal to:

$$t_1 = t_0 + \tau \quad [9]$$

From this, the model time constant can be found as:

$$\tau = t_1 - t_0 \quad [10]$$

7.4 QNET DC Motor Control Trainer

The DC Motor Control Trainer illustrates the fundamentals of DC motor controls using the NI ELVIS platform and NI LabVIEW. The experiment is designed to provide the understanding of how to use the virtual interface to take speed and voltage measurements of the responses. The components in the trainer are marker by the ID number in figure 7.3 and explained in the table 7.2.

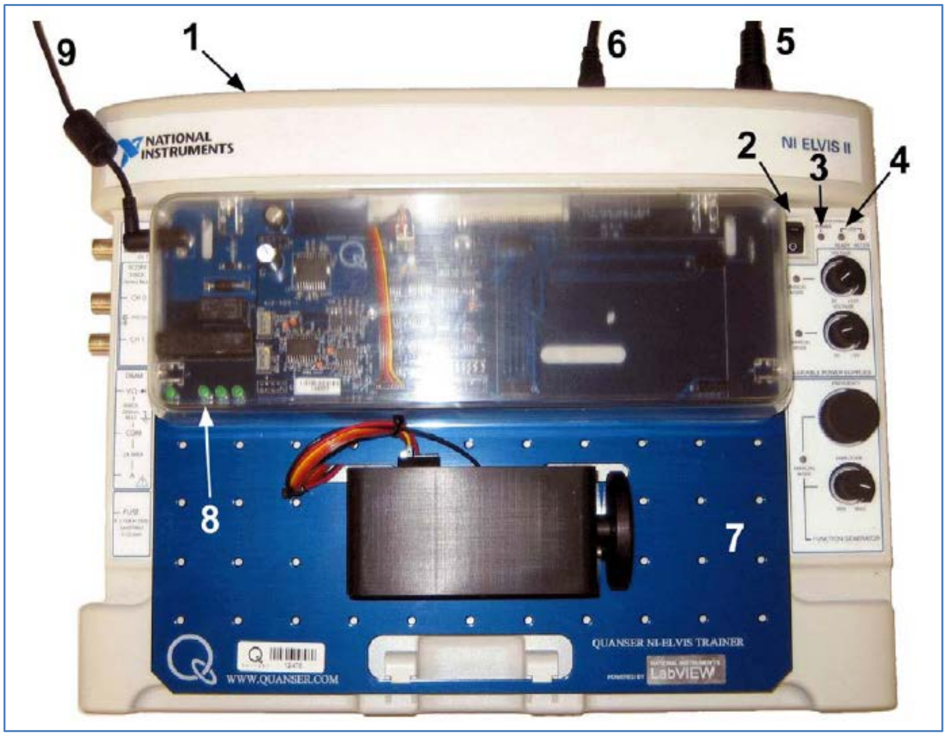


Figure 7.3: QNET DC motor control Trainer(DCMCT)

ID#	Description	ID#	Description
1	NI ELVIS II Trainer	6	Connection btw PC and ELVIS II
2	Prototyping Board Power switch	7	QNET DC Motor Control Trainer
3	Power LED	8	QNET Power LEDs
4	Ready LED	9	Power Cable for QNET
5	Power Cable for ELVIS II		

Table 7.2:ELVIS II and QNET Components

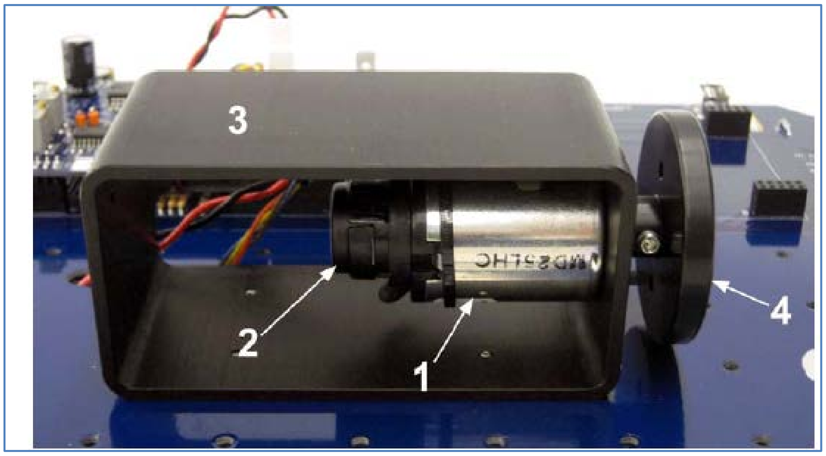


Figure 7.4: QNTDC motor components

The DCMCT components in the Figure 7.4 are marked by an ID number and explained in table 7.3

ID #	Description
1	DC motor
2	High resolution encoder
3	Motor metal chamber
4	Inertial Load

Table 7.3: DCMCT Component nomenclature

The DCMCT Modeling VU, runs the DC motor in open-loop and plots the corresponding speed and input voltage responses. This VI can be used to take speed and voltage measurements of the responses and runs a simulation of the DC motor in parallel. The table 7.4 describes the main elements of the QNET_DCMCT Model virtual instrument front panel.

Speed ω_m	Motor output speed numeric display rad/sec
Current I_m	Motor armature current numeric display A
VoltageV_m	Motor input voltage numeric display
K	Motor model steady-state gain input box rad/(V.s)
τ	Motor model time constant input box
Sampling rate	Sets the sampling rate of VI
Stop	Stops the LABVIE VI from running

Table 7.4: Nomenclature of QNET-DCMCT Modeling VI

7.5Procedure

<u>Identification of system parameters</u>

1. Open the QNET_DCMCT_Modeling.vi.
2. Ensure the correct Device is chosen as shown in figure 7.5

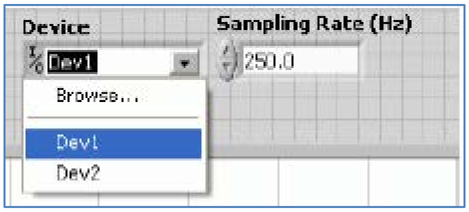


Figure 7.5: Selecting correct device

3. Run the QNET_DCMCT_Modeling.vi. The DC motor should begin spinning.
4. In the Signal Generator section set:
   Amplitude = 2.0 V

Frequency = 0.40 Hz
Offset = 3.0 V

5. Once you have collected a step response, click on the Stop Button to stop running the VI.
6. Select the Measurement Graphs tab to view the measured response.
7. Use the responses in the speed (rad/s) and Voltage (V) graphs to compute the steady-state gain of the DC motor and fill out the values in table 7.5. See the Bumptest Method section for details on how to find the steady-state gain from a step response. Finally you can use the Graph Palette for zooming functions and the Cursor Palette to measure data.

Description	***Symbol***	***Value***	***Unit***
Steady-state motor speed	ω_{ss}		rad/s
Initial step motor speed	ω_0		rad/s
Input step amplitude	A_v		V
Measured steady-state gain using bumptest	$K_{a,b}$		$rad(V.s)$

Table 7.5: Finding steady-state gain using bumptest

8. Based on the Bumptest Method, find the time constant. Make sure you complete table 5.6.

Description	***Symbol***	***Value***	***Unit***
Decay speed	$\omega_{ss}(t_1)$		rad/s
Initial step time	t_0		s
Decay step time	t_1		s
Measured time constant using bumptest	$\tau_{a,b}$		s

Table 7.6: Finding time constant using bumptest

9. Enter the steady-state gain and time constant values found in this section in table 5.7. These are called the bumptest model parameters.

Description	***Symbol***	***Value***	***Unit***
Open-Loop steady state Gain	$K_{a,b}$		$rad(V.s)$
Open-Loop Time Constant	$\tau_{a,b}$		s

Table 7.7: Bumptest Modeling

Model Validation

When the modeling is complete it is validated by running the model and the actual process in open loop. That is, open loop voltage is fed to both the model and the actual device such that both the simulated and measured response can be viewed on the same scope. The model can then be adjusted to fit the measured motor speed by fine-tuning the modeling parameters.

1. Open the QNET_DCMCT_Modeling.vi.
2. Ensure the correct Device is Chosen.
3. Run the QNET_DCMCT_Modeling.vi. You should hear the DC motor begin running.
4. In the Signal Generator section set:
   Amplitude = 2.0 V

Frequency = 0.40 Hz
Offset = 3.0 V

5. In the Model Parameters section of the VI, enter the bumptest model parameters, K and τ. The blue simulation should match the red measured motor speed more closely.**How well does your model represent the actual system? If they do not match, name one possible source of discrepancy**

6. Tune the steady-state gain, K, and the time constant τin the model Parameters section so the simulation matches the actual system better. Enter both the bumptest and tuned model parameters in table 7.8.

Description	***Symbol***	***Value***	***Unit***
Bumptest Modeling			
Open-Loop steady state Gain	$K_{a,b}$		$rad(V.s)$
Open-Loop Time Constant	$\tau_{a,b}$		s
Tuned Model Parameters			
Open-Loop steady state Gain	$K_{a,v}$		$rad(V.s)$
Open-Loop Time Constant	$\tau_{a.v}$		s

Table 7.8:QNET DCMCT modeling results summary

EXPERIMENT8

QNET DC Motor Speed Control

Objective:

- Design a PI controller to regulate the speed of the DC motor

8.1 DC Motor Speed Control

The experiment is designed in order to provide full understanding of PI controllers. The aim is to give you the knowledge to simulate and validate the controllers.

The speed of the DC motor is controlled using proportional integral control system. The PI control also includes set point weight. The transfer function representing the DC-motor speed voltage relation is used to design the PI controller.

The mathematical model of a DC motor has developed and its physical parameters are identified in previous experiment. Once the model is verified it is used to design a proportional-integral, or PI, controller that must meet certain given specifications.

The block diagram of the closed-loop system is shown in Figure 8.1.

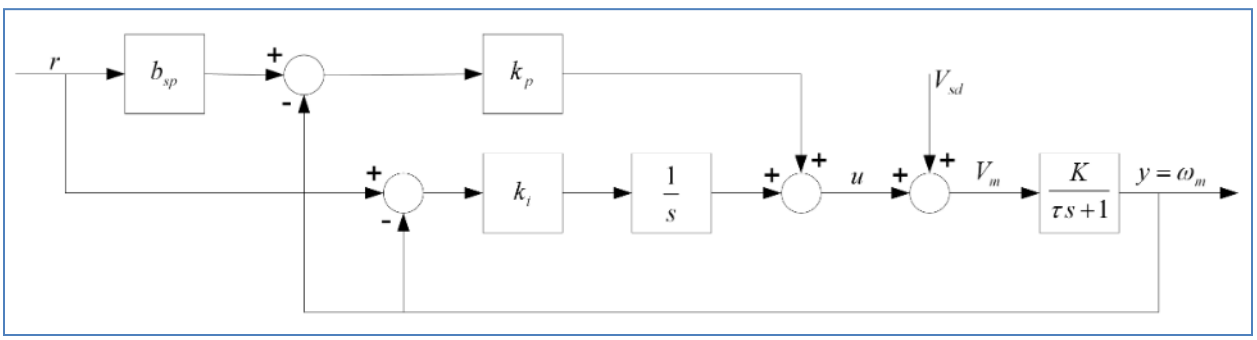


Figure 8.1: DC Motor PI closed-loop block diagram

The transfer function representing the DC motor speed-voltage relation in Equation 1 is used to design the PI controller. The input-output relation in the time-domain for a PI controller with set-point weighting is

$$u = k_p\left(b_{sp}r - y\right) + \frac{k_i(r-y)}{s} \qquad [1]$$

Where k_p is the proportional gain, k_i is the integral gain, and the bsp is the set point weight. The closed loop transfer function from the speed reference, r, to the angular motor speed output, ω_m, is

$$G_{\omega,r}(s) = \frac{K(k_P b_{sp} s + k_i)}{\tau s^2 + (K k_P + 1)s + K k_i} \qquad [2]$$

The standard second-order transfer function has the form

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} \quad [3]$$

Where ω_n is the natural undamped frequency and ζ is the damping ratio.

By comparing the characteristic equations of 2 and 3, we can find the expressions for control gains k_p and k_i.

$$k_p = \frac{-1+2\zeta\omega_n\tau}{K} \quad [4]$$

$$k_i = \frac{\omega_n^2\tau}{K} \quad [5]$$

8.2 Transient Characteristics of Second Order Systems

The closed loop system shown in Figure 8.1 represents a second-order system which is verified from equation 8.2. The properties of second-order systems response depend on the values of ω_n and ζ. Consider when a second order system, as shown in Equation3, is subjected to a step input given by

$$R(s) = \frac{R_0}{s} \quad [6]$$

with a step amplitude of $R_0 = 1.5$. The system response to this input is shown in Figure 8.2.

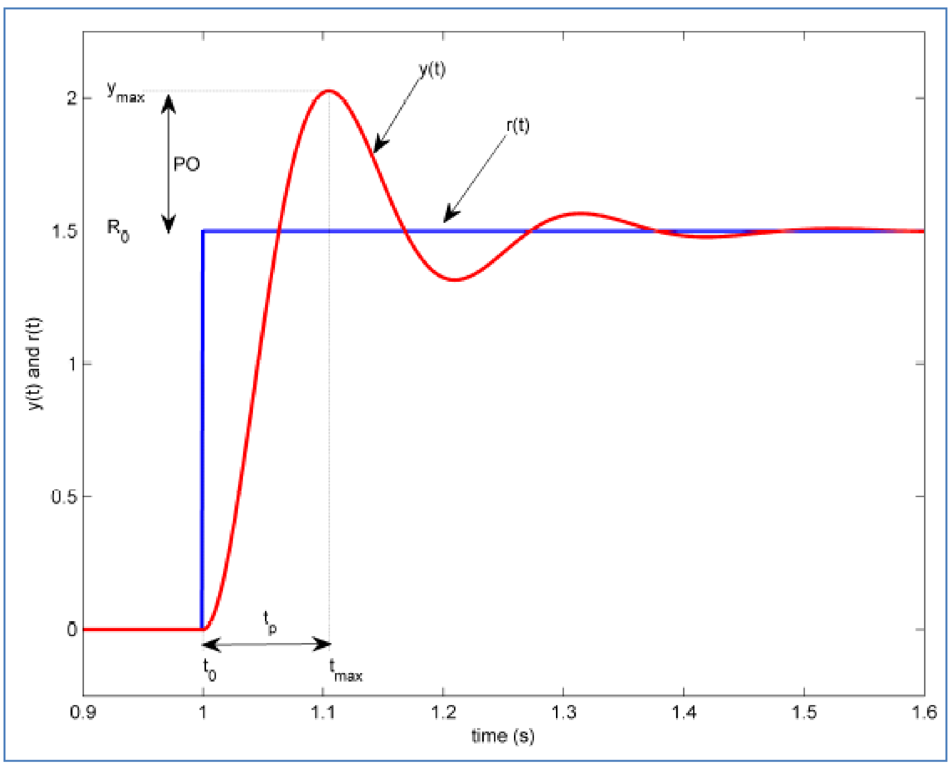


Figure 8.2: Standard second-order step response

The maximum value of the response is denoted by the variable y_{max} and it occurs at a time t_{max}. For a response similar to Figure 8.2, the percent overshoot is found using

$$PO = \frac{100\,(y_{max} - R_0)}{R_0}, \text{ here } y_{ss} = R_0 \qquad [7]$$

From the initial step time, t_0, the time it takes for the response to reach its maximum value is

$$t_p = t_{max} - t_0 \qquad [8]$$

This is called the peak time of the system.

In second order system, the amount of overshoot depends solely on the damping ratio parameter and it can be calculated using equation

$$PO = 100e^{\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)} \qquad [9]$$

The peak time depends on both the damping ratio and natural frequency of the system and it can be derived that the relationship between them is

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} \qquad [10]$$

The damping ratio affects the shape of the response while the natural frequency affects the speedof the response.

8.3 QNET DC Motor Speed VI:

The table 8.1 describes the main elements of the QNET_DCMCT Speed virtual instrument front panel

Label/Parameter	***Description***	***Unit***
Speed ω_m	Motor output speed numeric display	rad/s
CurrentI_m	Motor armature current numeric display	A
VoltageV_m	Motor input voltage numeric display	V
DisturbanceV_{sd}	Apply simulated disturbance voltage	V
k_p	Controller proportional gain input box	$V.s/rad$
k_i	Controller integral gain input box	V/rad
b_{sp}	Controller set point weight input box	

Table 8.1: Nomenclature of QNET-DCMCT Speed Control VI

8.4Procedure

8.4.1 Qualitative PI Control

1. Open the QNET_DCMCT_Speed_Control.vi.
2. Ensure the correct Device is chosen.
3. Run the QNET_DCMCT_Speed_Controlvi. The motor should begin rotating.

4. In the *Signal Generator* section set:
   Signal type = 'square wave'
   Amplitude = 25.0 rad/s
   Frequency = 0.40 Hz
   Offset = 100.0 rad/s

5. In the *Control Parameters* section set:
   k_p = 0.050 V.s/rad
   k_i = 1.00 V/rad
   bsp = 0.00

6. Examine the behavior of the measured speed, shown in red, with respect to the reference speed, shown in blue, in the *Speed (rad/s)* scope. Explain what is happening.

7. Increment and decrement kp by steps of 0.005 V.s/rad.
8. Look at the changes in the measured signal with respect to the reference signal. Explain the performance difference of changing kp.

9. Set k_p to 0 V.s/rad and k_i to 0 V/rad. The motor should stop spinning.
10. Increment the integral gain, k_i , by steps of 0.05 V/rad. Vary the integral gain between 0.05 V/rad and 1.00 V/rad.
11. Examine the response of the measured speed in the Speed (rad/s) scope and compare the result with k_i is set low to when it is set high. Explain what is happening?

12. Stop the VI by clicking on the Stop button.

8.4.2 PI Control According to Specifications

1. Using the equations 8.9 and 8.10 , Record the expected peak time, tp, and percentage overshoot, PO in table 8.1, given the following Speed Lab Designs (SLD) specifications:
   ζ= 0.75
   ω_n = 16.0 rad/s

Description		***Symbol***	***Value***	***Unit***
SLD specifications	Natural frequency	ω_n	16.0	rad/s
	Damping ratio	ζ	0.75	
Peak time		t_p		s
Percentage overshoot		PO		%

Table 8.2: Expected peak time and overshoot

2. Calculate the proportional, k_p, and the integral, k_i, control gains according to the model parameters (found in modeling section, Exp#7) and SLD specifications and write the calculated values in table 8.2.

Description		***Symbol***	***Value***	***Unit***
SLD specifications	Natural frequency	ω_n	16.0	rad/s
	Damping ratio	ζ	0.75	
Steady-state model gain		K		$rad/(V.s)$
Model time constant		τ		s
Proportional gain		k_p		$V.s/rad$
Integral gain		k_i		V/rad

Table 8.3: PI speed control design

3. Run the QNET_DCMCT_Speed_Control.vi. The motor should begin spinning.
4. In the Signal Generator set
   Signal type ='square wave'
   Amplitude = 25.0 rad/s
   Frequency = 0.40 Hz
   Offset = 100.0 rad/s
5. In the Control Parameters section, enter the SLD PI control gains found in table 8.2 and make sure bsp = 0.00.
6. Stop the VI when you obtain two sample cycles by clicking on the Stop button.
7. Capture the measured SLD speed response. Make sure you include both the Speed (rad/s) and the control signal Voltage (V) scopes.
8. Measure the peak time and percentage overshoot of the measured SLD response. Are the specifications satisfied?

9. Write down the effect of increasing the specification ζ have on the measured speed response in Table 8.3? How about on the control gains?

Description	***Symbol***	***Behaviour***	***Unit***
Peak time	t_p		s
Percent overshoot	PO		%
Proportional gain	k_p		$V.s/rad$
Integral gain	k_i		V/rad

Table8.4: Effect of increasing damping ratio specification in speed control

10. Write down the effect of increasing specification ω_n have on the measured speed response and the generated control gains in Table 8.4.

Description	***Symbol***	***Behaviour***	***Unit***
Peak time	t_p		s
Percent overshoot	PO		%
Proportional gain	k_p		$V.s/rad$
Integral gain	k_i		V/rad

Table8.5: Effect of increasing natural frequency specification in speed control

8.4.3 Effect of Set-Point Weight

1. Run the QNET_DCMCT_Speed_Controlvi. The motor should begin rotating.
2. In the Signal Generator section set:
   Signal type = 'square wave'
   Amplitude = 25.0 rad/s
   Frequency = 0.40 Hz
   Offset = 100.0 rad/s
3. In the Control Parameters section set:

$$k_p = 0.050\ V.s/rad$$

$k_i = 1.00\ V/rad$
$bsp = 0.00$

4. Increment the set-point weight parameter bsp in steps of 0.05. Vary the parameterbetween 0 and 1.
5. Examine the effect that raising bsp has on the shape of the measured speed signal in the Speed (rad/s) scope. Explain what the set-point weight parameter is doing.

6. Stop the VI by clicking on the Stop button

8.4.4. Tracking Triangular Signals

1. Run the QNET_DCMCT_Speed_Controlvi. The motor should begin rotating.
2. In the Signal Generator section set:
   Signal type = 'triangular wave'
   Amplitude = 50.0 rad/s
   Frequency = 0.40 Hz
   Offset = 100.0 rad/s
7. In the Control Parameters section set:

$k_p = 0.020\ V.s/rad$
$k_i = 0.00\ V/rad$
$bsp = 1.00$

3. Compare the measured speed and the reference speed. Explain why there is a tracking error

4. Increase ki to 0.1 V/rad and examine the response. Vary ki between 0.1 V/rad and 1.0 V/rad.
5. What effect does increasing ki have on the tracking ability of the measured signal? Explain using the observed behavior in the scope.

6. Stop the VI by clicking on the Stop button.

Post Lab

a) What do you understand by inertia, friction and dampening?

b) What kind of controller is used to control the speed of the motor? Give reasons to support your answer.

c) Explain the purpose of proportional, integral and derivative gains and how do they affect the speed of the motor.

EXPERIMENT 9

QNET DC Motor Position Control

Objective:

- Design a PD controller to regulate the speed of the DC motor

9.1DC Motor Position Control

The aim of this is to make you understand the system, commission the system, run and evaluate it.Control of motor position is a natural way to introduce the benefits of derivative action. In this experiment performance of proportional-derivative controller is observed.

The mathematical model of a DC motor has developed and its physical parameters are identified previously. Once the model is verified it is used to design a proportional-derivative, or PD, controller that must meet certain given specifications.

The block diagram of the closed-loop system is shown in Figure 7.1.

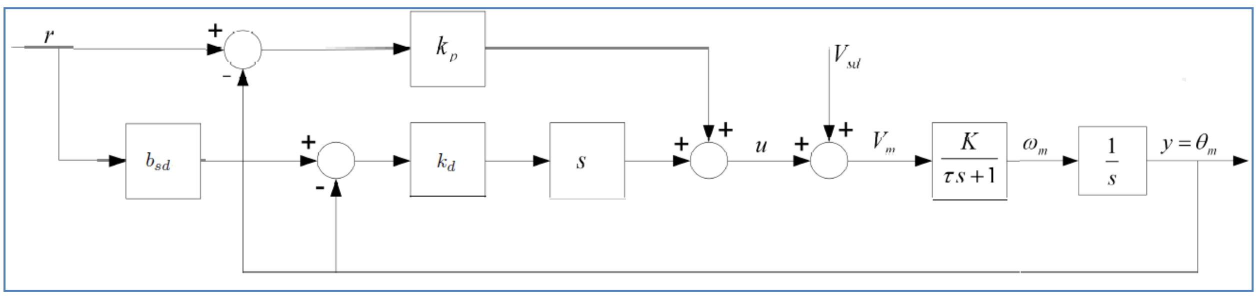


Figure 9.1: DC Motor PD closed-loop block diagram

The transfer function representing the DC motor angular position-voltage relation in Equation 9.1 is used to design the PD controller. The input-output relation in the time-domain for a PD controller is

$$u = k_p(r - y) + k_d(b_{sd}r - y) \qquad [1]$$

Where k_p is the proportional gain, k_d is the derivative gain. Combining the position process model

$$\frac{\theta_m(s)}{V_m(s)} = \frac{K}{s(\tau s+1)} \qquad [2]$$

with the PD control equation 7.1 gives the closed loop transfer function of the motor position system

$$G_{\theta,r}(s) = \frac{K(k_p+b_{sd}k_d s)}{\tau s^2+(1+Kk_d)s+Kk_p} \qquad [3]$$

Similarly, to the speed control experiment, the standard characteristic function shown in equation 3 can be achieved by setting the proportional gain to

$$k_p = \frac{\omega_n^2 \tau}{K} \qquad [4]$$

and the derivative gain to

$$k_d = \frac{-1+2\zeta\,\omega_n \tau}{K} \qquad [5]$$

9.2QNET DC Motor Speed VI:

The table 9.1summarized the main elements of the QNET_DCMCT positionVI front panel

Label/Parameter	*Description*	*Unit*
Position θ_m	Motor output position numeric display	rad
CurrentI_m	Motor armature current numeric display	A
VoltageV_m	Motor input voltage numeric display	V
DisturbanceV_{sd}	Apply simulated disturbance voltage	V
k_p	Controller proportional gain input box	$V.s/rad$
k_d	Controller derivative gain input box	$V.s/rad$

Table9.1: Nomenclature of QNET-DCMCTposition Control VI

9.3 Procedure

9.3.1 Qualitative PD Control

1. Open the QNET_DCMCT_Position _Control.vi.
2. Ensure the correct Device is chosen.
3. Run the QNET_DCMCT_Speed_Controlvi. The motor should begin rotating.
4. In the *Signal Generator* section set:
   Amplitude = 2.00 rad
   Frequency = 0.40 Hz
   Offset = 0.00 rad
5. In the *Control Parameters* section set:
   k_p= 2.00 V.s/rad
   k_i = 0.00 V/rad
   kd = 0.00 V/rad.
6. Change the proportional gain, k_p, by steps of 0.25 V/rad. Try the following gains k_p= 0.5, 1, 2, and 4 V/rad.
7. Examine the behavior of the measured position (red line) with respect to the reference position (blue line) in the *Position* (*rad*) scope. Explain what is happening.

8. Describe the steady-state error to a step input.

9. Increment the derivative gain, kd, by steps of 0.01 V.s/rad.
10. Look at the changes in the measured position with respect to the desired position. Explain what is happening.

9.3.2 PD Control according to Specifications

1. Using the equations in 8.9 and 8.10 (Experiment 8)calculate and record the expected peak time, t_p, and percentage overshoot, PO *in Table 9.1*, given
   Zeta = 0.60
   w0 = 25.0 rad/s
   p0 = 0.0

Description	***Symbol***	***Value***	***Unit***
Natural frequency specification	ω_n	25.0	rad/s
Damping ratio specification	ζ	0.6	
Peak time	t_p		s
Percentage overshoot	PO		%

Table9.1: Expected peak time and overshoot

2. Calculate the proportional, kp, and derivative, kd, control gains according to the model parameters found in previous experiment and the required specifications and write down theses values in Table 9.2

Description	***Symbol***	***Value***	***Unit***
Natural frequency specification	ω_n	25.0	rad/s
Damping ratio specification	ζ	0.6	
Steady-state model gain	K		$rad/(V.s)$
Model time constant	τ		s
Proportional gain	k_p		$V.s/rad$
Derivative gain	k_d		V/rad

Table9.2: PD speed control design

3. Run the QNET_DCMCT_Position_Control.vi. You should see the DC motor rotating back and forth.
4. In the Signal Generator section set:

Amplitude = 2.00 rad
Frequency = 0.4. Hz
Offset = 0.00 rad

5. In the control Parameters section set the PD gains found in Table 9.2.
6. Measure the peak time and percentage overshoot of the measured position response. Are the specifications satisfied? If they are not, then given on possible reason why there would be discrepancy.

7. Write down the changing the specification *zeta* have on the measured position response and the generated control gains in Table 9.3

Description	***Symbol***	***Behaviour***	***Unit***
Peak time	t_p		s
Percent overshoot	PO		%
Proportional gain	k_p		$V.s/rad$
Derivative gain	k_d		V/rad

Table9.3: Effect of increasing damping ratio specification in position control

8. Write down the effect of increasing the specification ω_n have on the measured position response and the generated control gains in Table 9.4

Description	***Symbol***	***Behaviour***	***Unit***
Peak time	t_p		s
Percent overshoot	PO		%
Proportional gain	k_p		$V.s/rad$
Derivative gain	k_d		V/rad

Table9.4: Effect of increasing natural frequency specification in position control

9. Stop the VI by Clicking on the *Stop* button.

Post Lab

a) What kind of controller is used to control the position of the motor? Give reasons to support your answer.

b) Explain the purpose of proportional, integral and derivative gains and how do they affect the position of the motor. You can write the equations to support your answer

EXPERIMENT 10

QNET Rotary Pendulum Trainer

Objective:

Rotary pendulum system is chosen for you to understand a task-based controller. The experiment begins by modeling the system and determines strategies to dampen the oscillations of the system.Furthermore, it makes you understand the relationship between stability and inertia of the system.

10.1 Modeling

The QNET-ROTPENT Simple Modeling runs the DC motor connected to the pendulum arm(as shown in Figure 10.1) in open-loop and plots the corresponding pendulum arm and link angles as well as the applied input motor voltage.

Figure 10.1(Pendulum)

10.1.1. Dampening

1. Open the QNET_ROTPENT_Simple_Modeling.vi.
2. Ensure the correct Device is chosen as shown in Figure 10.2.

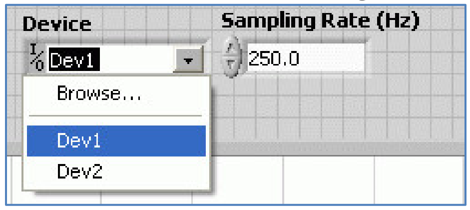


Figure10.2

3. Run the QNET_ROTPENT_Simple_Modeling.vi.
4. Hold the arm of the rotary pendulum system stationary and manually perturb the pendulum.
5. While still holding the arm, examine the response of Pendulum Angle (deg) in the Angle (deg) scope. This is the response from the pendulum system.
6. Repeat Step 3 above and release the arm after several swings.
7. Examine the Pendulum Angle (deg) response when the arm is not fixed. This is the response from the rotary pendulum system. Given the response from these two systems pendulum and rotary pendulum – which converges faster towards angle zero? Why does one system dampen faster than the other?

8. Stop the VI by clicking on the Stop button.

10.1.2. Friction

1. Run the QNET_ROTPENT_Simple_Modeling.vi.
2. In the Signal Generator section set:

    Amplitude = 0.00 V
    Frequency = 0.25 Hz
    Offset = 0.00 V111

3. Change the Offset in steps of 0.10 V until the pendulum begins moving. Record the voltage at which the pendulum moved.

4. Repeat Step 3 above for steps of -0.10 V.

5. Enter the positive and negative voltage values needed to get the pendulum moving in Table 10.1. Why does the motor need a certain amount of voltage to get the motor shaft moving?

Description	Symbol	Value	Unit
Positive Coulomb Friction Voltage	V_{fp}		V
Negative Coulomb Friction Voltage	V_{fn}		V

Table10. 1

6. Stop the VI by clicking on the Stop button.

10.1.3. Moment of Inertia

1. Using references QNET User Manual and QNET Practical Control Guide, calculate the moment of inertia acting about the pendulum pivot.

2. Run the QNET_ROTPENT_Simple_Modeling.vi
3. In the Signal Generator section set:
   Amplitude = 1.00 V
   Frequency = 0.25 Hz
   Offset = 0.00 V

4. Click on the Disturbance toggle switch to perturb the pendulum and measure the amount of time it takes for the pendulum to swing back-and-forth in a few cycles (e.g. 4 cycles).
5. Find the frequency and moment of inertia of the pendulum using the observed results. See QNET Practical Control Guide to see how to calculate the inertia experimentally and make sure you fill the following Table10.2.

Description	Symbol	Value	Unit
Cycles	n_{cyc}		
Duration	Δt		s
Frequency	f		Hz
Pendulum moment of inertia	$J_{p,exp}$		$kg.m^2$

Table 10.2

6. Compare the moment of inertia calculated analytically in step 1 and the moment of inertia found experimentally. Is there a large discrepancy between them?

7. Stop the VI by clicking on the Stop button.

Post Lab

a) What is rotary inverted pendulumsystem?

b) Define dampening and friction?

EXPERIMENT 11

QNET-ROTPENT Balance Control Design

Objective:

The experiment illustrates one of the important methods for finding the parameters of control strategies. It will give you deeper understanding of LQR technique that is suitable for finding the parameters of the balance controller.

11.1 Model Analysis

1. Open the QNET_ROTPENT_Control_Design.vi.
2. Run the QNET_ROTPENT_Control_Design.vi.
3. Select the Symbolic Model tab.
4. The Model Parameters array includes all the rotary pendulum modeling variables that are used in the state-space matrices A, B, C, and D.
5. Select the Open Loop Analysis tab.
6. This shows the numerical linear state-space model and a pole-zero plot of the open-loop inverted pendulum system. What do you notice about the location of the open-loop poles? Recommended: In the Model Parameters section, it is recommended to enter the pendulum moment of inertia, Jp, determined experimentally in previous experiment.

7. In the Symbolic Model tab, set the pendulum moment of inertia, Jp, to 1.0e-5 kg.m^2.
8. Select the Open Loop Analysis tab. How did the locations of the open-loop poles change with the new inertia? Enter the pole locations of each system with a different moment of inertia in the following table 11.1. Are the changes of having a pendulum with a lower inertia as expected?

Description	Symbol	Value	Unit
System w/ Jp = 1.7e-4	p_0		rad/s
	p_1		rad/s
	p_2		rad/s
	p_3		rad/s
System w/ Jp = 1.00e-5	p_0		rad/s
	p_1		rad/s
	p_2		rad/s
	p_3		rad/s

Table 11. 1

9. Reset the pendulum moment of inertia, Jp, back to 1.77e-4.
10. Stop the VI by clicking on the Stop button.

11.2 Control Design and Simulation

1. Open the QNET_ROTPENT_Control_Design.vi.
2. Select the Simulation tab.
3. Run the QNET_ROTPENT_Control_Design.vi.
4. In the Signal Generator section set:
   Amplitude = 45.0 deg
   Frequency = 0.20 Hz
   Offset = 0.0 deg
5. Set the Q and R LQR weighting matrices to the following:
   Q (1, 1) = 10, i.e. set first element of Q matrix to 10.
   R = 1.00
6. Changing the Q matrix generates a new control gain.
7. The arm reference (in red) and simulated arm response (in blue) are shown in the Arm (deg) scope. How did the arm response change? How did the pendulum response change in the Pendulum (deg) scope?

8. Set the third element in the Q matrix to 0, i.e. Q (3, 3) = 0.
9. Examine and describe the change in the Arm (deg) and Pendulum (deg) scope.

10. By varying the diagonal elements of the Q matrix, design a balance controller that adheres to the following specifications:
Arm peak time less than 0.75 seconds: $tp \leq 0.75\ s$
Motor voltage peak less than ± 12.5 V: $|Vm| \leq 12.5\ V$
Pendulum angle less than 10.0 degrees: $|\alpha| \leq 10.0\ deg$

11. Enter the Q and R matrices along with and control gain used to meet the specifications in Table 11.2.

Description	***Symbol***	***Value***	***Unit***
Q(1,1)	$Q_{1,1}$		
Q(2,2)	$Q_{2,2}$		
Q(3,3)	$Q_{3,3}$		
Q(4,4)	$Q_{4,4}$		
R	R		
K(1)	$k_{p,\theta}$		V/rad
K(2)	$k_{p,\alpha}$		V/rad
K(3)	$k_{d,\theta}$		V.s/rad
K(4)	$k_{d,\alpha}$		V.s/rad

Table 11.2

Post Lab

a) How is inertia related to stability? What does the pole-zero plot tells you about stability of the inverted pendulum system?

b) How is the arm response of pendulum related to proportional and derivative gain?

EXPERIMENT 12

Design of Light Intensity Control Using Hardware Based PID Controller(Open Ended Lab)

Problem Statement

Design and implement a hardware based PID controller which will control the intensity of light based upon the reference value that has been provided by the user. You are available with the LM741 operational amplifiers, BJT and some assorted resistors and capacitors. For sensing the intensity of light, you have Light Dependent Resistor (LDR) and you have to think carefully that how can you sense the intensity variations in terms of voltage. As an output device, you have a filament-based DC bulb which can bear a maximum voltage of 5V and draws almost 500mA. Display your results on oscilloscope.

EXPERIMENT 13

Root Locus Analysis

Objective:

1. To plot the root locus for a given transfer function of the system using MATLAB
2. To use the root locus techniques to analyze the effect of loop gain upon system's transient response and stability

13.1 Design and analysis using Root Locus

Root locus, a graphical presentation of the closed-loop poles as a system parameter is varied. Root loci are used to study the effects of varying feedback gains on closed-loop pole locations. In turn, these locations provide indirect information on the time and frequency response.
The root locus of an open-loop transfer function $KG(s)H(s)$ is the plot of all possible closed loop pole locations with proportional gain K, which varies from zero to infinity.

In order to plot the root locus entire s-plane is required to be traversed but this is very tedious job. We can plot the root locus using computer program e.g. MATLAB. The open loop transfer function of the system shown in Figure 13.1 is KG(s)H(s).

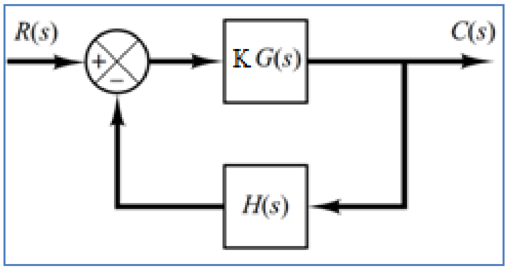


Figure 13.1: Closed Loop System with gain K

The command used to plot root locus in MATLAB is

sys = tf(num,den)
rlocus(sys) //calculates and plots the root locus of the open-loop SISO model sys

Root locus of a given transfer function can be plotted using following set of commands in MATLAB:

```
clc;
closeall;
clearall;
num=[2 1]; % Define numerator of Open Loop Transfer Function
den=[2 -3 5]; % Define denominator of Open Loop Transfer Function
sys=tf (num,den)
rlocus (sys) % Draw root locus
title ('Root Locus')
```

Note: rlocus(A,B,C,D) command can be used when system is defined in State Space.

Closed-loop poles can be determined at any value of gain (K) using following MATLAB command. Gain K can also be provided as vector for which close loop poles are to be determined.

```
sys=tf (num,den)
K=0.5; % Specify the value of Gain
[R]=rlocus (sys, K)  %Returns matrix R of complex poles for given
value of K
[R]=rlocus(A,B,C,D,K)%For system represented in State Space
```

Value of gain (K) and respective closed-loop poles can also be determine by locating any point on the root locus. Following MATLAB command will pop-up the root locus of the given system, any point can be located to find corresponding value of gain and complex poles.

```
sys=tf(num,den)
rlocus (sys)
[K,p]=rlocfind(sys) %Gives the Gain K and its corresponding poles for a
point at root locus
```

Constant Damping Ratio (ζ)and Constant Natural Frequency (ω_n)Loci

Constant damping ratio (ζ) lines are radial lines having constant slope passing though the origin and constant natural frequency (ω_n) are the circles of constant radius.

Following MATLAB command is used to draw lines of constant damping ratio and circles of constant natural frequency as shown in Figure 13.2:

```
sys=tf (num,den) %Open Loop Transfer Function
rlocus (sys) % Draw Root Locus
zeta=[0.3, 0.5, 0.8]; %Values of Zeta
wn=[1, 4, 7]; %Values of Natural Frequency Draw
sgrid(zeta,wn);%Draw lines(boundaries)of Zeta and wn
```

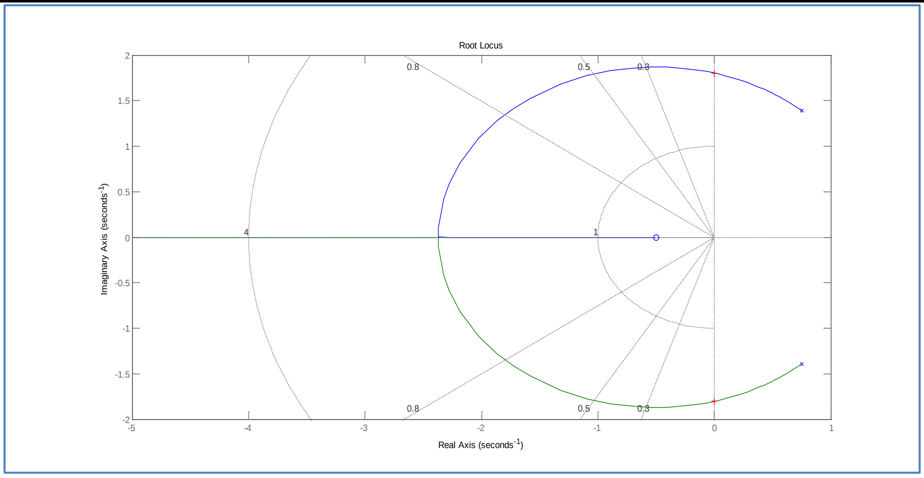


Figure13.2: RL with ζ and ω_n lines

Constant zeta and constant natural frequency lines can be omitted by usingempty brackets in sgrid command.
For example: **sgrid(z, [])**omits the constant ω_n lines.

Exercise 1

Consider the unity feedback system in figure 13.3:

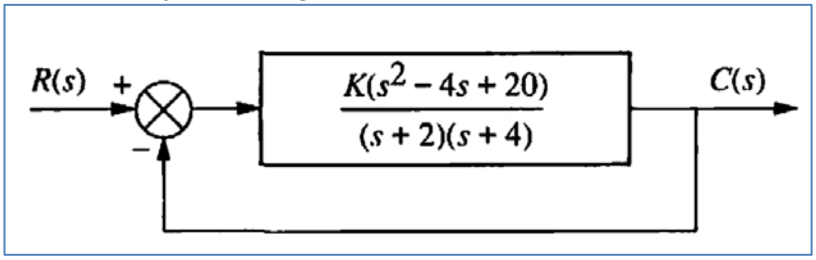


Figure 13.3: Control system

Use MATLAB to find following:

a. Location of closed loop poles at K=0 and K=∞?
b. Find the break points
c. The point and gain where the locus crosses the $j\omega$-axis
d. Find close loop pole at gain K=1
e. The range of K within which the system is stable
f. Find the value of closed loop poles which yield 0.45 damping ratio. Also determine the value of gain K at this point.

Exercise 2

Consider the unity feedback system where:

$$G(s) = \frac{K(s+5)(s+4)}{s^2}$$

Using MATLAB perform the following tasks.

a. Determine the closed loop poles for K=0 and K=∞

b. Find the value of K that yields closed loop critically damped poles.

c. The point and gain where the locus crosses the $j\omega$-axis

d. Find the range of gain K for which system is stable

e. Find the value of gain K and location of closed loop poles that yields a damping ratio of 0.75

f. Find the point (closed loop pole) on root locus that yields zeta 0.667 and natural frequency 3rad/sec.

Exercise 3

For the following unity feedback system

$$G(s) = \frac{K(s+3)}{s(s+2)(s^2+4s+5)}$$

Using MATLAB perform the following task.

a. Determine the closed loop poles for K=0 and K=∞

b. Find the break points and value of K at break point.

c. Find the location of closed loop poles and the value of K at which system is marginally stable?

d. Find the value of K for which dominant closed loop poles have damping ratio 0.35

Note: Save the MATLAB graphs in your PC in order to get verified from your instructor.

Post Lab

In figure 13.4, the block diagram of the closedloop control of the linearized magnetic levitation system is shown. (Galvao, 2003)

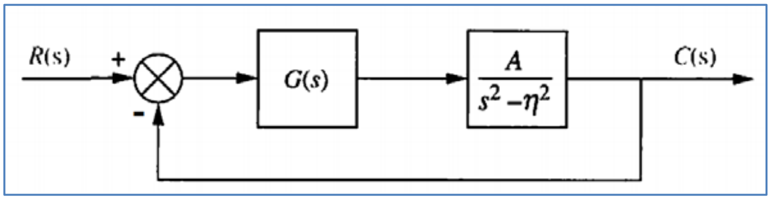


Figure 13.4: Linearized magnetic levitation system block diagram

Assuming A = 1300 and η= 860, draw the root locus andfind the range of K for closed-loop stability when:

A. $G(s) = K$
B. $G(s) = \frac{K(s+200)}{s+1000}$

Note:In your report include Root locus diagrams, any gain K you found using the rlocfindcommand and whether this gain is making system stable/unstable/marginally stable.

EXPERIMENT 14

Compensator Design using Root Locus

Objective:

To successfully design a compensator to meet the transient response and steady state error performance specifications

The experiment is designed to understand the graphical procedure for determining the stability of a control system based on root locus. MATLAB Control system tool box contains two root locus design GUI, sisotool and rltool. The main aim of this experiment is to use SISO toolbox to understand and design the compensators using root locus design.

14.1 SISO Toolbox Tutorial

SISOTOOL opens a SISO Design GUI for interactive compensator design. This GUI allows you to design a single-input/single-output (SISO) compensator using root locus.

By default, the SISO Design Tool:

- Opens the Control and Estimation Tools Manager with a default SISO Design Task node.
- Opens the Graphical Tuning editor with root locus and open-loop Bode diagrams.
- In current architecture, compensatorC is placed in the forward path in series with the plant G.
- Assumes the prefilter, F, and the sensor, H, are unity gains. Once you specify G and H, they are *fixed* in the feedback structure.

To make the plant model and start the SISO Design Tool, at the MATLAB prompt type

>>sisotool

An empty SISO design tool opens as shown in figure 14.1.

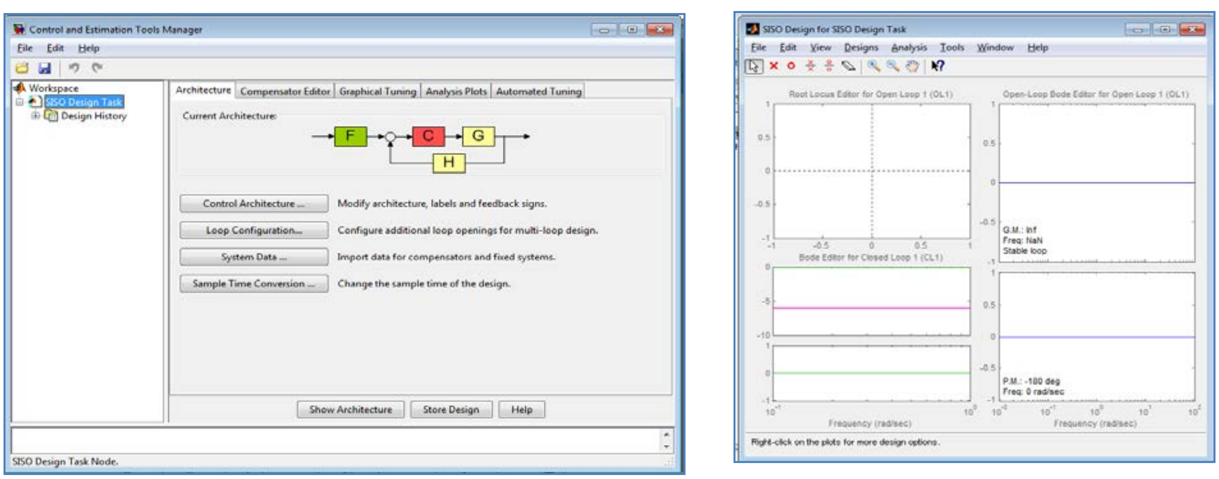


Figure 14.1: SISO Design Tool

The default control architecture is shown in the current architecture of SISO design tool.

Example 1:

In the Control System shown in the figure 14.2, G(s) is a proportional controller, K.

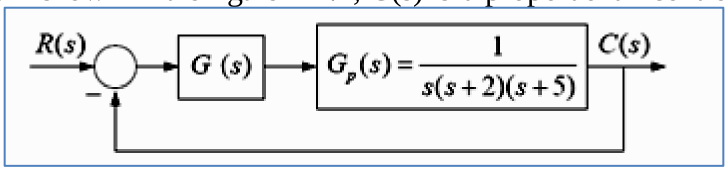


Figure 14.2:Control System

Using sisotool, determine the following:

a) Range of K for system stability

b) Value of K for the complex dominant poles damping ratio 0.6. For this value of K obtain the step response and time domain specifications.

To start the SISO Design Tool, at the MATLAB prompt type

```
>>Gp = tf(1,   [ 1 7 10 0])
>>sisotool
```

An empty SISO Design Tool opens.The SISO Design Tool by default assumes that the compensator is in the forward path. Select System Data to Import Model. This opens the System Data dialog box, which is shown in figure 14.3.

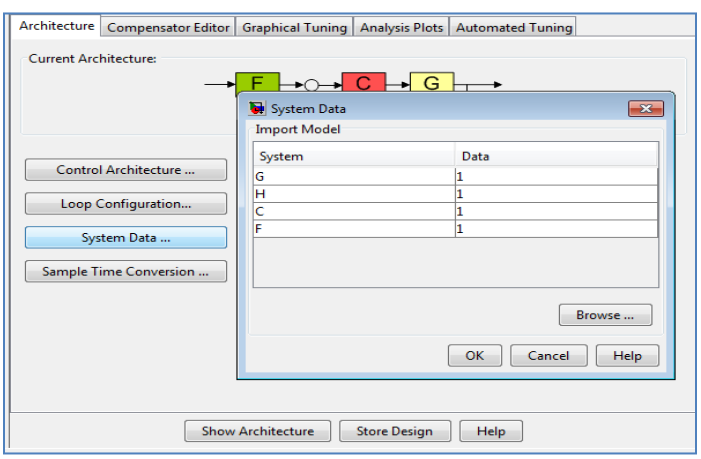


Figure 14.3: Import Model

All the available models will appear in the Model Listbox: The Blocks are coded as F (A preamplifier), G (Plant model) H (Sensor) with default values of 1. By using Browse option import transfer function Gpfor plant model from workspace as shown in figure 14.4.

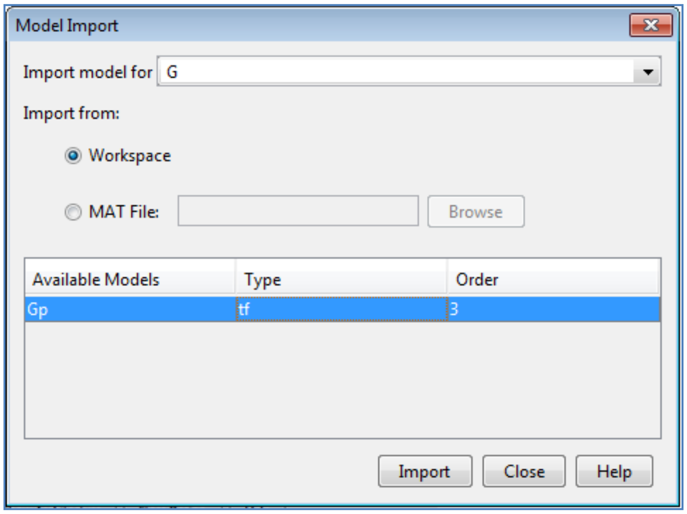


Figure 14.4: Import model from workspace

The root locus and open-loop Bode diagrams will be displayed. The red squares on the loci represent the closed loop poles corresponding to gain set point. Also the gain margin and phase margin are shown on the bode plot.From Analysis select Other Loop Responses. This opens the Analysis plots window as shown in Figure 14.5. Select Step plot type for closed loop r to y and click on Show Analysis Plot.

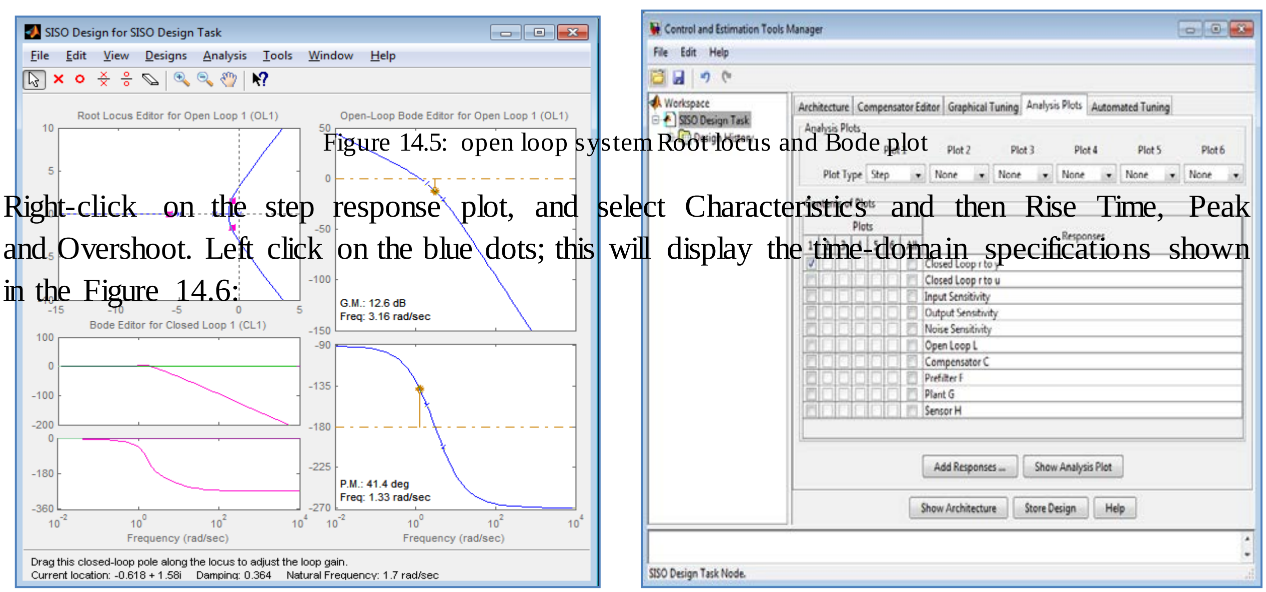


Figure 14.5: open loop system Root locus and Bode plot

Right-click on the step response plot, and select Characteristics and then Rise Time, Peak and Overshoot. Left click on the blue dots; this will display the time-domain specifications shown in the Figure 14.6:

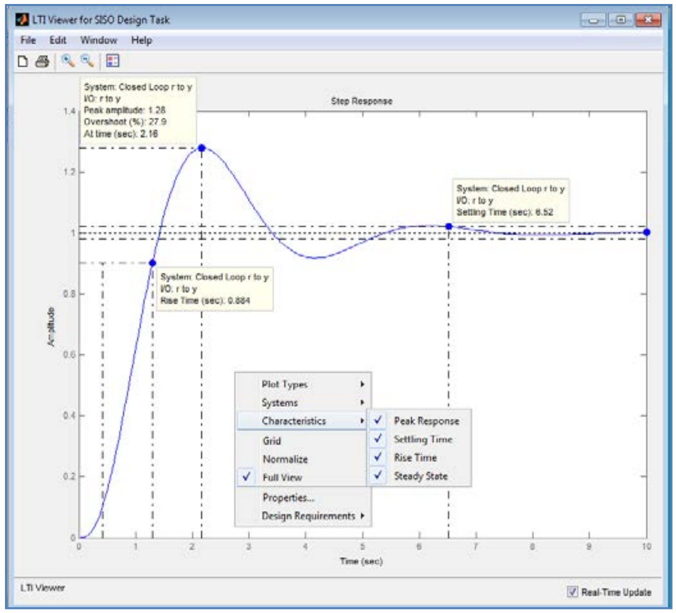


Figure 14.6: Step response of closed loop system

Example 2:

For the unity feedback system, $G_p(s) = \frac{1}{s(s+2)(s+5)}$ design a phase lead compensator

$$G_c(s) = \frac{K_c(s+z_c)}{(s+p_c)}$$

For the following time domain specifications:

- Dominant poles damping ratio=0.707
- Dominant poles settling time=2sec

On the matlab prompt type

```
>>Gp=tf([1],[1 7 10 0])
```

```
>>sisotool
```

An empty SISO design tool opens. Select **Import Model** under the **System Data**. This opens the **Import System Data** dialog box. Select Gp and place it in the G field. Click **OK.** The root locus and open loop bode diagrams are displayed in the plot regions.

To place a pair of pole and zero on your diagram at a damping ratio of 0.707, right-click on the root locus and select **New**from**Design Requirements**. This opens the Design Requirements editor as shown in figure 14.7.

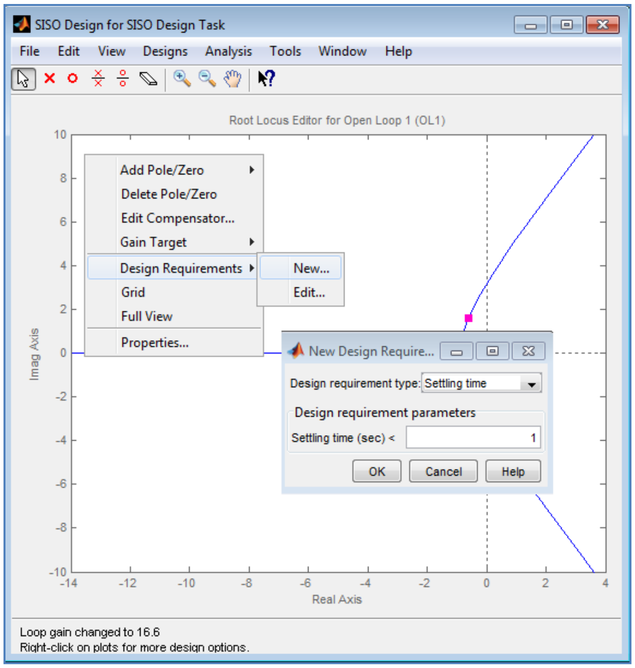


Figure 14.7:Design requirements editor

In the damping ratio field enter 0.707. This results in a pair of rays at the desired slope as shown in figure 13.8. To add a settling time constraint, reopen the **New Design Requirements** window and choose settling time from the pull-down menu and set the field to 2.

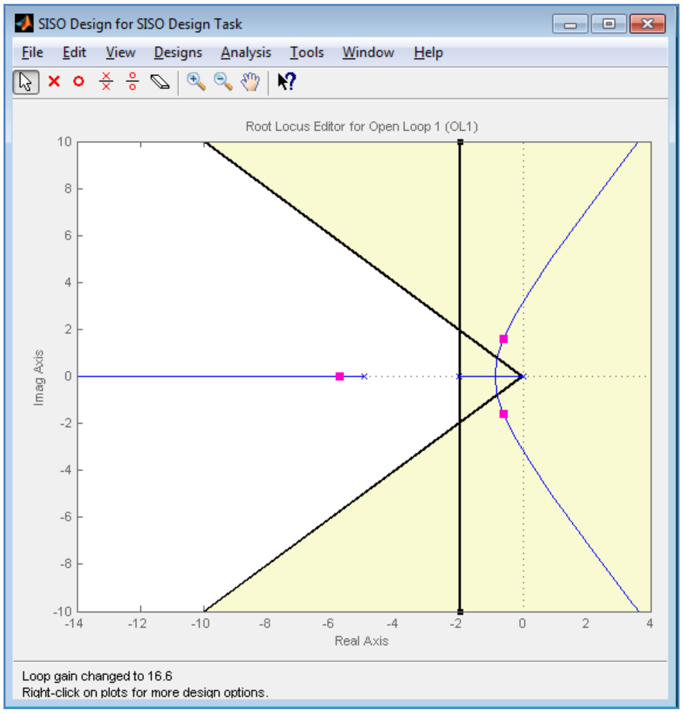


Figure14.8: After setting the design requirements

In **Designs** tab select **Edit Compensator**.I0n the **Edit Compensator** type -1.75 for a Real Zero to place the compensators zero at -1.75. Also, add a real pole, on the real axis to the left of the pole with the largest magnitude. Move the closed-loop pole up the loci towards the intersection of ζ and τ constraints. You can turn on the Mouse zoom and place a square in the area containing the controller zero, pole and the closed loop pole. Adjust the pole position on the real axis and at the same time hold the left mouse and drag the closed-loop pole until the pole is placed at the desired location as appears on the bottom panel. To view the compensator transfer function in pole zero format, select appropriate compensators format from SISO Tool Preferences menu as shown in figure 14.9.

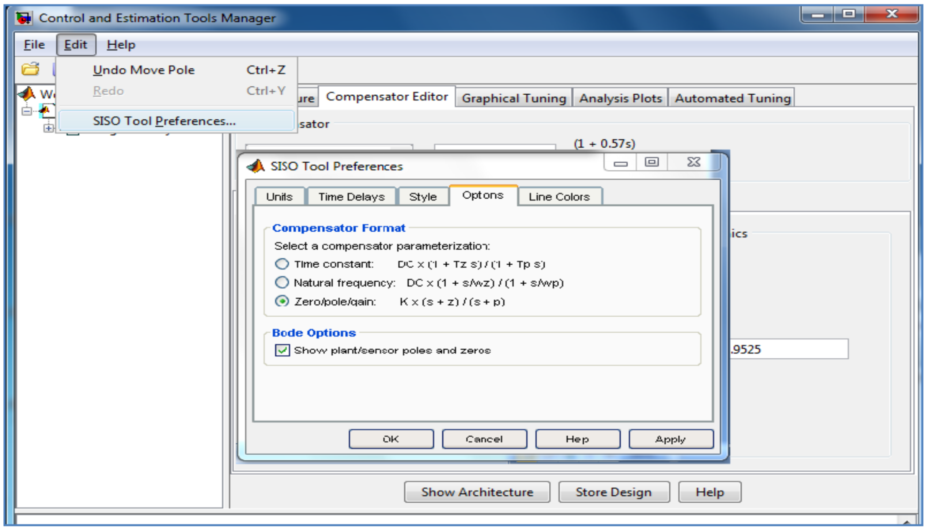


Figure 14.9: Compensator format

Also verify the systems time response using step response plot and Record these values in Table in 14.1.

Observations:

Transfer function of Lead Compensator	
Transfer function of Compensated System	
Damping ratio	
Settling time	

Table 14.**1:** Compensated systemand its performance

Exercise 1: Reevaluate the system's performance when the lead compensator's zero is placed at -2 in example 2. Record the values in Table 14.2

Transfer function of Lead Compensator	
Transfer function of Compensated System	
Damping ratio	
Settling time	

Table 14.2: Compensated system and its performance

Exercise 2: A unity feedback system has open loop transfer function

$$G(s)H(s) = \frac{4}{s(s+2)}$$

It is desired that dominant closed loop poles provide damping ratio=0.5 and have an un-damped natural frequency 4rad/sec. Velocity error constant is required to be 40.

Verify that only gain adjustment cannot meet these objectives.

Design 1

a) Design a lead compensator using SISO design tool to meet the objectives, when compensator zero is placed at -1.
b) Using GUI, Record the peak overshoot and settling time of the lead-compensated system in Table 14.3.

Transient response	***Value***	***Units***
Peak Overshoot		%
Settling Time		*s*
Peak Time		*s*

Table 14.3: Transient response analysis of compensated system

Design 2

a) Design a lead compensator using SISO design tool to meet the objectives, when compensator zero is placed at -2.
b) Using GUI, Record the peak overshoot and settling time of the lead-compensated system in Table 14.4.

Transient response	***Value***	***Units***
Peak Overshoot		%
Settling Time		*s*
Peak Time		*s*

Table 14.4: Transient response analysis of compensated system

Which design is preferable in exercise 2, in order to achieve the performance specifications and why?

POST LAB

Use the following plant:

$$P(s) = \frac{8.96}{0.00147s^2 + 0.01455s + 1}$$

Simulate its step response. Using a Lead Compensator, try to get the settling time to less than 0.5 sec (approx. 0.1 sec) while keeping the %OS lesser than 10%. Justify your design in the report.

Appendix A: Lab Evaluation Criteria

Labs with projects

1. Class Participation 10%
2. Lab Work 40%
3. Quiz (2) 20%
4. Project 30%
    a. Project Demonstration 20%
    b. Project Report 5%
    c. Project Quiz 5%

Labs without projects

1. Class Participation 10%
2. Lab Work 40%
3. Quiz (2) 20%
4. Lab Final 30%
    a. Lab Final (Practical) 20%
    b. Lab Final (Written) 10%

Appendix B: Safety around Electricity

In all the Electrical Engineering (EE) labs, with an aim to prevent any unforeseen accidents during conduct of lab experiments, following preventive measures and safe practices shall be adopted:

- Remember that the voltage of the electricity and the available electrical current in EE labs has enough power to cause death/injury by electrocution. It is around 50V/10 mA that the "cannot let go" level is reached. "The key to survival is to decrease our exposure to energized circuits."
- If a person touches an energized bare wire or faulty equipment while grounded, electricity will instantly pass through the body to the ground, causing a harmful, potentially fatal, shock.
- Each circuit must be protected by a fuse or circuit breaker that will blow or "trip" when its safe carrying capacity is surpassed. If a fuse blows or circuit breaker trips repeatedly while in normal use (not overloaded), check for shorts and other faults in the line or devices. Do not resume use until the trouble is fixed.
- It is hazardous to overload electrical circuits by using extension cords and multi-plug outlets. Use extension cords only when necessary and make sure they are heavy enough for the job. Avoid creating an "octopus" by inserting several plugs into a multi-plug outlet connected to a single wall outlet. Extension cords should ONLY be used on a temporary basis in situations where fixed wiring is not feasible.
- Dimmed lights, reduced output from heaters and poor monitor pictures are all symptoms of an overloaded circuit. Keep the total load at any one time safely below maximum capacity.
- If wires are exposed, they may cause a shock to a person who comes into contact with them. Cords should not be hung on nails, run over or wrapped around objects, knotted or twisted. This may break the wire or insulation. Short circuits are usually caused by bare wires touching due to breakdown of insulation. Electrical tape or any other kind of tape is not adequate for insulation!
- Electrical cords should be examined visually before use for external defects such as: Fraying (worn out) and exposed wiring, loose parts, deformed or missing parts, damage to outer jacket or insulation, evidence of internal damage such as pinched or crushed outer jacket. If any defects are found the electric cords should be removed from service immediately.
- Pull the plug not the cord. Pulling the cord could break a wire, causing a short circuit.
- Plug your heavy current consuming or any other large appliances into an outlet that is not shared with other appliances. Do not tamper with fuses as this is a potential fire hazard. Do not overload circuits as this may cause the wires to heat and ignite insulation or other combustibles.
- Keep lab equipment properly cleaned and maintained.
- Ensure lamps are free from contact with flammable material. Always use lights bulbs with the recommended wattage for your lamp and equipment.
- Be aware of the odor of burning plastic or wire.
- ALWAYS follow the manufacturer recommendations when using or installing new lab equipment. Wiring installations should always be made by a licensed electrician or other qualified person. All electrical lab equipment should have the label of a testing laboratory.
- Be aware of missing ground prong and outlet cover, pinched wires, damaged casings on electrical outlets.
- Inform Lab engineer / Lab assistant of any failure of safety preventive measures and safe practices as soon you notice it. Be alert and proceed with caution at all times in the laboratory.

- Conduct yourself in a responsible manner at all times in the EE Labs.
- Follow all written and verbal instructions carefully. If you do not understand a direction or part of a procedure, ASK YOUR LAB ENGINEER / LAB ASSISTANT BEFORE PROCEEDING WITH THE ACTIVITY.
- Never work alone in the laboratory. No student may work in EE Labs without the presence of the Lab engineer / Lab assistant.
- Perform only those experiments authorized by your teacher. Carefully follow all instructions, both written and oral. Unauthorized experiments are not allowed.
- Be prepared for your work in the EE Labs. Read all procedures thoroughly before entering the laboratory. Never fool around in the laboratory. Horseplay, practical jokes, and pranks are dangerous and prohibited.
- Always work in a well-ventilated area.
- Observe good housekeeping practices. Work areas should be kept clean and tidy at all times.
- Experiments must be personally monitored at all times. Do not wander around the room, distract other students, startle other students or interfere with the laboratory experiments of others.
- Dress properly during a laboratory activity. Long hair, dangling jewelry, and loose or baggy clothing are a hazard in the laboratory. Long hair must be tied back, and dangling jewelry and baggy clothing must be secured. Shoes must completely cover the foot.
- Know the locations and operating procedures of all safety equipment including fire extinguisher. Know what to do if there is a fire during a lab period; "Turn off equipment, if possible and exit EE lab immediately."